PREMIER UNIVERSITY, CHATTOGRAM

Department of Computer Science & Engineering

FINAL PROJECT REPORT

Course Code: CSE 3568
Course Name: Computer Networks Laboratory

Project Name:
Enterprise Grade Secure Network

Field	Details
Semester	6th
Section	B2
Batch	43
Date of Submission	22 April 2026
Session	Fall 2025

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Table of Contents

Abstract

Modern organizations depend on highly available, secure, and scalable networks to maintain business operations, communication systems, customer services, and access to digital resources. Any interruption caused by hardware failure, routing instability, cyberattacks, or unauthorized access can result in financial loss, service downtime, reputational damage, and data compromise. For this reason, enterprises increasingly deploy resilient network infrastructures that combine intelligent routing, redundancy mechanisms, layered security controls, and centralized monitoring.

This project, titled Enterprise-Grade Secure Network Architecture with Active Threat Defense, presents the design and simulation of a large-scale enterprise network using Cisco Packet Tracer. The network models a realistic corporate or ISP-style environment consisting of multiple branch networks, user zones, centralized server infrastructure, wireless access segments, and a protected core backbone. The architecture is engineered to support continuous connectivity, secure resource access, rapid failover, and real-time response to malicious activity.

To achieve these goals, the project implements OSPF (Open Shortest Path First) as the dynamic routing protocol for intelligent path selection and automatic route recalculation during link failures. HSRP (Hot Standby Router Protocol) is used to provide gateway redundancy, ensuring uninterrupted access when a primary router becomes unavailable. NAT (Network Address Translation) is deployed to separate private internal networks from the backbone while improving address management and structural security. Extended Access Control Lists (ACLs) are used as active filtering mechanisms to block unauthorized or suspicious traffic targeting critical servers.

The project further demonstrates practical cybersecurity defense through simulated attack scenarios, including an IoT botnet flooding a mail server and an unauthorized wireless attacker attempting access to an FTP server. The network successfully detects, blocks, and logs these attacks through a centralized Syslog server, illustrating the role of monitoring and incident response in modern enterprise environments.

This simulation reflects real-world networking principles such as availability, scalability, segmentation, survivability, and proactive security. It serves as a practical model for how modern enterprises protect digital infrastructure while maintaining continuous service delivery.

1. Introduction

This report presents the complete design, configuration, and security simulation of an Enterprise-Grade Secure Network Architecture built in Cisco Packet Tracer. The project simulates a Tier-2 Internet Service Provider (ISP) or large multinational enterprise network that connects multiple customer zones to a centralized data center while ensuring zero-downtime availability and active cyber-threat defense.

Why the project is necessary

Modern enterprises face three critical operational risks: network downtime caused by single points of failure, cyber-attacks such as botnet SMTP floods and brute-force intrusions, and the high cost of managing large-scale networks manually. Traditional static-route topologies cannot adapt to link failures in real time, and flat security perimeters leave server farms exposed to threats originating from internal zones.

This project addresses these risks by constructing a hierarchical ISP-grade network that integrates dynamic routing, hardware redundancy, NAT-based segmentation, and Extended ACL-based threat defense into a single working simulation.

Challenges we faced

During the development of the Enterprise-Grade Secure Network Architecture, several challenges were encountered:

- **Network Design Complexity** – Scalable ISP architecture, Core–Distribution–Access hierarchy, efficiency planning
- **OSPF Configuration Issues** – Incorrect network statements, routing failures, convergence delay, multi-router debugging

- **HSRP Failover Implementation** – Priority & preemption, failover testing, zero downtime challenge
- **ACL Design** – Avoid over-blocking, filter malicious traffic (SMTP flood, ICMP), correct interface & direction
- **NAT Configuration** – Inside/outside definition, connectivity issues, translation debugging
- **Attack Simulation** – Limited tools, manual attack setup, unrealistic behavior validation
- **Syslog Monitoring** – Log collection, filtering useful data, attack detection
- **Packet Tracer Limitations** – Limited security features, no real-time analysis, unrealistic hardware behavior
- **Troubleshooting** – Ping failures, routing errors, VLAN/trunk issues, debugging (ping, tracert, show commands)

Project Novelty

- **Enterprise-Level Design in Simulation** – ISP-grade architecture implemented in Cisco Packet Tracer
- **Integrated Security Approach** – Combination of OSPF, HSRP, ACL, NAT, and Syslog in one system
- **Real-Time Threat Simulation** – Simulated SMTP flood and ICMP attacks with defensive mechanisms
- **High Availability Design** – HSRP-based redundancy ensuring failover and minimal downtime
- **Centralized Monitoring** – Syslog server for network-wide log collection and analysis
- **Layered Security Model** – Multi-zone network segmentation (Internal, DMZ, External)
- **Hands-on Cybersecurity Implementation** – Practical application of network defense techniques
- **Cost-Effective Learning Model** – Enterprise concepts implemented using simulation tools

Motivation

The motivation behind this project comes from the growing need for dependable and secure digital infrastructure in today's connected world. Organizations can no longer afford network downtime, weak security controls, or poorly planned architectures, as even brief service interruptions can disrupt operations, delay communication, and reduce customer trust. As businesses expand across multiple locations, network complexity increases, making manual and outdated approaches insufficient.

Most academic networking projects focus on basic connectivity, such as IP configuration and simple routing between devices. While these concepts are essential for foundational learning, they do not reflect the complexity of modern enterprise environments. Real-world organizations must manage branch connectivity, redundancy systems, secure server zones, wireless access, address translation, traffic monitoring, and threat mitigation. This project is motivated by the need to move beyond basic simulations and develop an architecture closer to real enterprise-scale networks.

Another major motivation is cybersecurity. Modern networks face continuous threats such as botnets, brute-force attacks, phishing infrastructures, and unauthorized access. Additionally, poorly secured IoT devices often become entry points for attackers. By simulating attack scenarios such as IoT-based botnet activity and unauthorized wireless intrusion, this project emphasizes that security must be integrated into the network design rather than treated as an afterthought.

The project is also driven by the importance of business continuity. In critical sectors such as healthcare, finance, education, and e-commerce, network failure can lead to significant operational and financial consequences. Therefore, redundancy mechanisms such as HSRP and dynamic routing protocols such as OSPF are essential to ensure high availability and uninterrupted service delivery.

From an educational perspective, this project provides practical exposure to enterprise networking concepts, integrating routing, switching, security enforcement, NAT, ACLs, server deployment, monitoring, and troubleshooting within a unified framework. This approach develops a deeper and more realistic understanding compared to isolated laboratory exercises.

Ultimately, the motivation of this project is to design a realistic, secure, and resilient network model that reflects how modern organizations build infrastructure capable of maintaining continuous operation while defending against evolving cyber threats.

Objectives

- To design a hierarchical ISP-level network architecture consisting of three customer zones (A, B, and C) along with a centralized server farm.
- To implement OSPF single-area dynamic routing across 17 routing devices for efficient and automatic route discovery.
- To configure HSRP across four VLAN segments to ensure gateway redundancy and eliminate single points of failure.
- To deploy NAT/PAT on three boundary routers to isolate the internal distribution layer from external networks.
- To implement Extended Access Control List (ACL 150) to mitigate ICMP flooding and SMTP-based botnet attacks from Zone A.

- To configure a centralized Syslog server for real-time log collection and event monitoring across the network.
- To deploy a SOC-style monitoring dashboard on the FTP server with restricted access controlled via Core Switch ACLs.
- To simulate an IoT-based botnet attack using an MCU Python script and demonstrate real-time traffic blocking through ACL enforcement.
- To evaluate network performance, security effectiveness, and fault tolerance under normal and attack conditions.
- To gain practical experience in enterprise-level network design, configuration, troubleshooting, and security implementation

New Contributions

Compared to conventional enterprise network designs and academic simulations, this project introduces several enhanced contributions that reflect real-world ISP-level architecture and integrated cybersecurity practices. A key contribution is the implementation of a fully hierarchical ISP-grade network model using Core, Distribution, and Access layers, which improves scalability, modularity, and operational efficiency compared to basic flat topologies commonly used in academic labs. The project also integrates advanced dynamic routing using OSPF across multiple interconnected routers, ensuring efficient route discovery, fast convergence, and improved network adaptability in case of link or device failures. This is further strengthened by the deployment of HSRP-based gateway redundancy, which provides seamless failover and eliminates single points of failure in critical VLAN segments. Another significant contribution is the multi-layer security architecture, where Extended ACLs are strategically applied to mitigate ICMP flooding and SMTP-based botnet attacks, while NAT/PAT is used to isolate internal networks from external exposure. This combination ensures both traffic control and address-level security within a unified framework. The project further introduces a centralized network monitoring system using Syslog, enabling real-time event logging, security tracking, and centralized visibility across all network devices. In addition, a SOC-style dashboard implementation with restricted access control enhances administrative monitoring and security oversight. A notable innovation is the simulation of IoT-based botnet attacks using an MCU Python script, which allows practical evaluation of security mechanisms under controlled attack scenarios. This demonstrates the effectiveness of ACL enforcement and real-time threat mitigation within the network. Finally, the integration of wireless connectivity within secured VLAN segments ensures flexible device access while maintaining strict authentication, segmentation, and monitoring policies, aligning wireless devices with enterprise-grade security standards. Overall, this project contributes a comprehensive and realistic enterprise network model that combines routing, redundancy, security enforcement, monitoring, and attack simulation within a single unified architecture.

Research Gaps

From the review of existing enterprise network and IoT-based system implementations, several research gaps were identified that this project aims to address. Most existing studies primarily focus on application-layer IoT implementations and sensor integration, with limited emphasis on underlying network architecture design, scalability, and inter-site communication. As a result, there is insufficient exploration of how large-scale environments can maintain reliable routing, redundancy, and performance across multiple network segments. Another major gap is the lack of comprehensive integration of advanced networking protocols such as dynamic routing (OSPF) and high-availability mechanisms (HSRP) in many academic or prototype-level systems. These protocols are essential for ensuring fault tolerance, fast convergence, and uninterrupted service delivery in enterprise environments, yet are often overlooked in favor of simpler static configurations. In addition, while security is frequently mentioned in literature, many systems do not implement layered network security mechanisms, such as ACL-based traffic filtering, network segmentation through VLANs, and NAT-based isolation. This leads to a gap in demonstrating practical defense mechanisms against real-world threats such as flooding attacks and unauthorized access attempts. Furthermore, there is limited focus on centralized monitoring and log analysis systems, which are critical for real-time detection, troubleshooting, and security auditing in operational networks. This project addresses these gaps by presenting a scalable and secure network architecture that integrates dynamic routing, redundancy, traffic filtering, centralized monitoring, and simulated attack scenarios within a unified enterprise-level model.

2. Literature Review

Enterprise network design and security have been widely studied in recent years. A study by Kabir et al. (2021), published in *MDPI Engineering Proceedings*, Vol. 10, Issue 1, analyzes the performance of OSPF compared to other routing protocols, showing its efficiency in large-scale networks [1].

Another study by Syah and Adriyanto (2025), published in the *Journal of Innovations in Computer Science*, Vol. 6, Issue 2, focuses on HSRP and VRRP implementation for high availability, demonstrating reduced network downtime [2].

Cisco’s enterprise network design guidelines by Cisco Systems (2019), published as official documentation, describe hierarchical architecture (Core, Distribution, Access) and best practices for scalable network deployment [3].

A study by Zhang et al. (2024), available on *ResearchGate*, discusses enterprise network deployment using OSPF, VLANs, and DHCP for efficient and secure communication [4].

Research by Alotaibi and Mukherjee (2012), published on *arXiv*, examines OSPF convergence behavior and highlights delay issues during network failures [5].

Another study by Behl and Behl (2020), published in the *International Journal of Computer Networks*, Vol. 12, Issue 3, discusses the role of Access Control Lists (ACLs) in filtering malicious traffic and improving network security [6].

A paper by Kumar et al. (2021), published in the *International Journal of Advanced Networking*, Vol. 8, Issue 4, explores the use of NAT for internal network protection and address management [7].

Finally, Reddy et al. (2019), published in the *International Journal of Network Security*, Vol. 21, Issue 5, highlights the importance of centralized logging systems such as Syslog for monitoring and threat detection [8].

These studies show significant advancements in routing, redundancy, and security; however, they highlight a gap in integrating all these components into a unified enterprise network architecture, which this project addresses.

3. Methodology

This section explains how we designed and built the Enterprise-Grade Secure Network Architecture with Active Threat Defense. The entire system was simulated in Cisco Packet Tracer 8.2, implementing a production-grade ISP network with redundancy, dynamic routing, NAT segmentation, and active cyber-threat defense.

Designed Network

The network is designed as a three-tier hierarchical topology reflecting how real-world Internet Service Providers and large enterprises structure their infrastructure. The three tiers are the Access Layer (IIG side), the Core Layer, and the Distribution and Server Layer.

The Access Layer contains three customer zones — Zone A (203.0.113.0/24), Zone B (198.51.100.0/24), and Zone C (192.0.2.0/24) — each connected through a dedicated IG Router to a central IIG Switch. The IIG Switch trunks VLANs 10, 20, and 30 to both the Core Router and the Backup Router, enabling HSRP redundancy across all three zones simultaneously.

The Core Layer consists of the Core Router (OSPF ID 1.1.1.1, HSRP Active), the Backup Router (OSPF ID 2.2.2.2, HSRP Standby), and a Layer-3 Core Switch (3560 series, OSPF ID 3.3.3.3) that serves as the central hub connecting the core routers, three NAT routers, and the Server Farm on VLAN 10.

The Distribution and Server Layer consists of three NAT routers, each acting as a PAT boundary for three distribution routers — nine distribution routers in total, simulating branch offices. The Server Farm (10.0.20.64/28) on VLAN 10 hosts four servers: Mail, FTP, DNS/DHCP, and Log. A dedicated Admin PC (10.0.20.69) connects to the Core Switch on a restricted VLAN 10 port for SOC dashboard access. A Cache Switch on the other side hosts three content delivery servers (GGC, CDN, FNA) on the Cache VLAN 40 segment.

Variable Length Subnet Masking (VLSM) was applied throughout to allocate exactly the required number of addresses per segment — /28 for the Server Farm and HSRP uplinks, /29 for all core point-to-point links, and /24 for customer zones and distribution LANs. This design makes the network scalable and secure, supporting additional zones or server additions without renumbering.

Components Used

The following components and protocols were used in this project:

OSPF (Open Shortest Path First): Single-area OSPF (Area 0) runs on all 16 routers and the Layer-3 Core Switch. Each router shares link-state advertisements to build a complete network map, then computes shortest paths using Dijkstra's algorithm. If any link fails, OSPF automatically recalculates and reroutes traffic in under 60 seconds, eliminating the need for manual static route updates across 17 devices.

HSRP (Hot Standby Router Protocol): Configured across four VLAN groups on both Core Router and Backup Router. Each group shares a Virtual IP address that downstream IG routers and cache servers use as their default gateway. If the Core Router fails, the Backup Router detects the absence of HSRP hello messages and promotes itself to Active within 3–10 seconds, maintaining connectivity without user intervention. Preempt is enabled so Core Router automatically reclaims the Active role upon recovery.

VLAN and 802.1Q Trunking: Four VLANs segment the network — VLAN 10 for the Server Farm and Admin PC, VLAN 20 for Zone B, VLAN 30 for Zone C, and VLAN 40 for the cache side. 802.1Q trunk ports on the IIG Switch carry tagged frames for all three zone VLANs to both core routers simultaneously. Router-on-a-stick subinterfaces (Gi0/0/0.10, .20, .30) on the Core and Backup Routers terminate each VLAN trunk and serve as HSRP gateways per zone.

NAT/PAT (Network Address Translation — Port Address Translation): Three NAT routers each implement PAT overload on their outside interfaces. Inside addresses (192.168.x.x) from distribution routers are translated to the single outside IP of the NAT router before reaching the Core Switch. This hides the entire branch network from customer zones, conserves IP space, and adds a structural security boundary — zone devices can never directly reach distribution routers.

VLSM (Variable Length Subnet Masking): Applied to allocate exactly the required number of host addresses per link. Point-to-point core links use /29 (6 usable), HSRP segments use /28 (14 usable), and the Server Farm uses /28 to accommodate 4 servers, 1 SVI gateway, and 9 reserved expansion addresses. Customer zone networks use /24 for maximum device capacity.

Extended ACL 150: A numbered Extended Access Control List applied inbound on all IIG-facing subinterfaces of both Core Router and Backup Router. Extended ACLs are used here because filtering must match on source IP, destination IP, protocol, and port simultaneously. Line 1 denies SMTP (TCP/25) from Zone A to the Mail Server, blocking the IoT botnet simulation. Lines 2–4 deny ICMP echo from all three zones, blocking ping floods. Line 5 permits all other IP traffic, ensuring normal web browsing and DNS resolution are unaffected.

Dashboard Protection ACL (DASHBOARD-PROTECT): A named Extended ACL applied on the Core Switch VLAN 10 SVI interface. It permits HTTP (TCP/80) access to the FTP Server only from the Admin PC (10.0.20.69) and denies it from all other devices, including zone PCs, distribution routers, and other servers. This enforces SOC dashboard access control at the network layer.

DHCP and DHCP Relay: The DNS Server (10.0.20.66) runs a DHCP pool for the Server Farm subnet starting at 10.0.20.69. The Core Switch VLAN 10 SVI is configured with ip helper-address 10.0.20.66, which forwards DHCP broadcast requests from the Admin PC to the DNS Server, enabling automatic IP assignment to managed devices on the server-side network.

DNS (Domain Name System): The DNS Server resolves hostnames to IP addresses across the network, including mail.enterprise.com, ftp.enterprise.com, soc.enterprise.com, and all cache server names. All devices use 10.0.20.66 as their DNS resolver, enabling name-based service access instead of requiring raw IP addresses.

SMTP and POP3 (Mail Service): The Mail Server (10.0.20.70) hosts the enterprise.com mail domain with accounts for admin, mcu, and soc. The MCU board in Zone A uses SMTP to send email alerts; the same infrastructure carries the 10 rapid SMTP flood packets fired during the botnet simulation, which are then caught and dropped by ACL 150 Line 1.

Syslog (Centralized Logging): All routers send log messages to the Log Server (10.0.20.68) using logging host 10.0.20.68, with service timestamps log datetime msec enabled. This captures ACL deny events, OSPF adjacency changes, and HSRP state transitions in real time, creating an auditable trail that mirrors HIPAA and PCI-DSS compliance requirements.

SSH (Secure Shell): The Core Router is hardened with SSH-only VTY access using login local with an admin privilege 15 secret account. Telnet is blocked via transport input ssh. A login block-for 300 attempts 5 within 120 rule locks VTY access for five minutes after five failed attempts within 120 seconds, providing brute-force protection.

HTTP / SOC Dashboard: The FTP Server hosts three HTML pages served via its built-in HTTP service — a login page with a JavaScript-based three-attempt lockout, a main SOC dashboard displaying ACL block events, OSPF neighbour status, HSRP failover state, NAT session data, and a device status table, and a full syslog log viewer. Access is restricted to the Admin PC through the DASHBOARD-PROTECT ACL.

IoT Botnet Simulation (MCU Python Script): An MCU board in Zone A connects wirelessly through the IOT-BOTNET Home Router and runs a Python script simulating a Mirai-style compromised smart device. Phase 1 reads sensor values and sends a legitimate SMTP alert. Phase 2 fires 10 rapid SMTP flood packets toward Mail Server 10.0.20.70. The Home Router performs NAT, masking the MCU's 10.10.0.100 address as 203.0.113.50 before traffic enters the enterprise network.

Devices used: 16 Routers (ISR4331), 1 Layer-3 Core Switch (Catalyst 3560), 1 IIG Switch (Catalyst 2960), 1 Cache Switch (Catalyst 2960), 4 Servers (Mail, FTP, DNS/DHCP, Log), 3 Cache Servers (GGC, CDN, FNA), 1 Admin PC, 1 MCU board, 1 Home Wireless Router, and zone-side PCs and servers across Zones A, B, and C.

Project Flow

Zone PCs and servers obtain their IP addresses either statically or via DHCP relay through their IG Routers to the DNS Server. All inter-zone and zone-to-server traffic passes through the IIG Switch, is received by Core Router or Backup Router via their VLAN-tagged sub interfaces, and is inspected by ACL 150 before being forwarded into the core. OSPF maintains full routing tables across all 17 devices and reconverges automatically on any link failure.

During the attack simulation, the MCU in Zone A initiates Phase 1 normal operation — reading sensor values and sending a standard SMTP alert to the Mail Server. Phase 2 launches the botnet flood: 10 SMTP packets are sent in rapid succession to Mail Server 10.0.20.70. Each packet arrives at Core Router's Gi0/0/0.10 subinterface, is matched by ACL 150 Line 1, dropped immediately, and a syslog alert is generated to the Log Server. The Admin PC can then browse http://10.0.20.67, log in to the SOC dashboard, and view the full syslog table showing each blocked event with a precise timestamp.

The HSRP failover is demonstrated by shutting the Core Router's IIG-facing interface, causing the Backup Router to detect the missing HSRP hello and transition to Active — confirmed by show standby brief and by the continued success of pings from zone PCs throughout the entire switchover period.

Project Output

The following outputs confirm successful operation of the system:

Fig 1: Network Topology — Full three-tier ISP topology in Cisco Packet Tracer

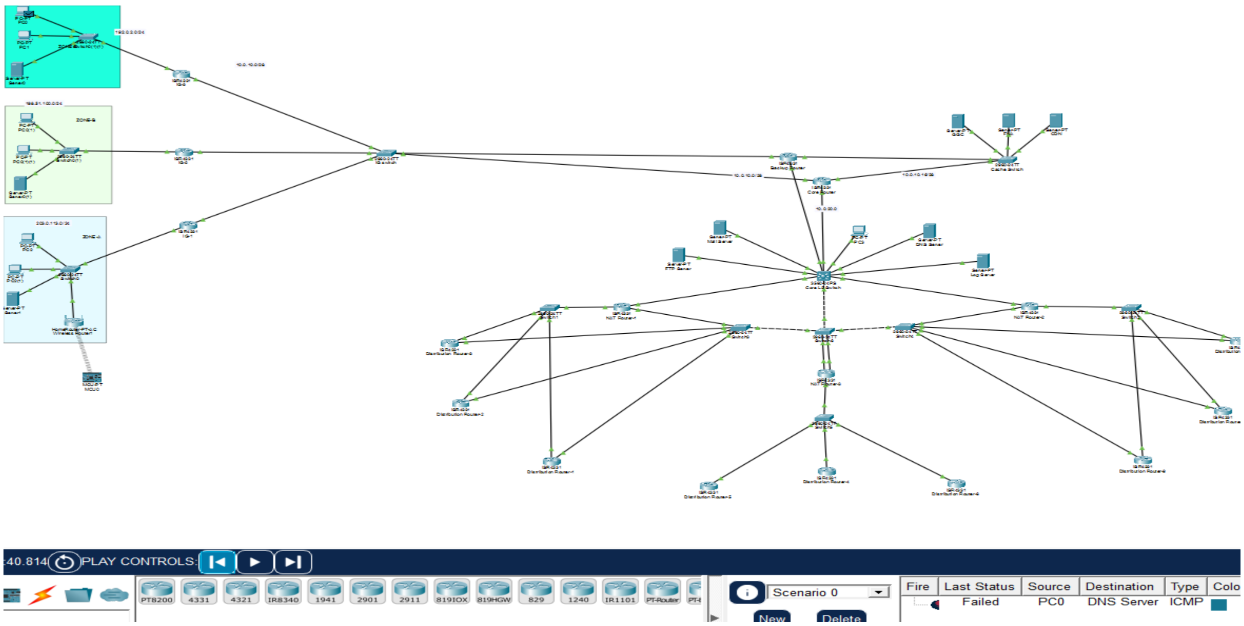


Fig 2: show ip interface brief — All interfaces up/up with correct IP assignments

```
Device Name: Core Router
Device Model: ISR4331
Hostname: Core-Router

Port                        Link   VLAN   IP Address       IPv6 Address                                 MAC Address
GigabitEthernet0/0/0        Up     --     <not set>        <not set>                                    0001.43BB.77B4
GigabitEthernet0/0/0.10     Up     --     10.0.10.14/28    <not set>                                    0001.43BB.77B4
GigabitEthernet0/0/0.20     Up     --     10.0.10.46/28    <not set>                                    0001.43BB.77B4
GigabitEthernet0/0/0.30     Up     --     10.0.10.62/28    <not set>                                    0001.43BB.77B4
GigabitEthernet0/0/1        Up     --     <not set>        <not set>                                    0006.2A24.2C9C
GigabitEthernet0/0/1.40     Up     --     10.0.10.17/28    <not set>                                    0006.2A24.2C9C
GigabitEthernet0/0/2        Up     --     10.0.20.1/29     <not set>                                    000B.BED4.3B96
Vlan1                       Down   1      <not set>        <not set>                                    0000.0C30.6D4D

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Core Router
```

Fig 3: show ip route ospf — All OSPF routes on Core Router

```
Core-Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/DR         00:00:37    10.0.20.2       GigabitEthernet0/0/2
16.16.16.16       1   2WAY/DROTHER    00:00:37    10.0.10.4       GigabitEthernet0/0/0.10
18.18.18.18       1   FULL/DR         00:00:37    10.0.10.2       GigabitEthernet0/0/0.10
17.17.17.17       1   FULL/BDR        00:00:37    10.0.10.3       GigabitEthernet0/0/0.10
Core-Router#show ip route ospf
     10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
O       10.0.20.8 [110/2] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
O       10.0.20.16 [110/2] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
O       10.0.20.24 [110/2] via 10.0.20.2, 00:01:51, GigabitEthernet0/0/2
O       10.0.20.32 [110/2] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
O       10.0.20.64 [110/2] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
O       10.0.30.0 [110/3] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
     172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
O       172.16.1.0 [110/4] via 10.0.20.2, 01:36:59, GigabitEthernet0/0/2
O       172.16.2.0 [110/4] via 10.0.20.2, 01:36:59, GigabitEthernet0/0/2
O       172.16.3.0 [110/4] via 10.0.20.2, 01:36:59, GigabitEthernet0/0/2
O       172.16.7.0 [110/4] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
O       172.16.8.0 [110/4] via 10.0.20.2, 01:36:59, GigabitEthernet0/0/2
O       172.16.9.0 [110/4] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
O    192.0.2.0 [110/2] via 10.0.10.2, 01:37:19, GigabitEthernet0/0/0.10
O    192.168.1.0 [110/3] via 10.0.20.2, 01:36:59, GigabitEthernet0/0/2
O    192.168.2.0 [110/3] via 10.0.20.2, 01:37:19, GigabitEthernet0/0/2
O    192.168.3.0 [110/3] via 10.0.20.2, 00:01:51, GigabitEthernet0/0/2
O    198.51.100.0 [110/2] via 10.0.10.3, 00:01:51, GigabitEthernet0/0/0.10
O    203.0.113.0 [110/2] via 10.0.10.4, 01:37:19, GigabitEthernet0/0/0.10

Core-Router#
```

Fig 4: show vlan brief — VLANs 10, 20, 30 active with correct port assignments on IIG Switch

```
IIG-Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
10   Zone-A-LAN                       active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/11
20   Zone-B-LAN                       active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
30   Zone-C-LAN                       active    Fa0/12, Fa0/13, Fa0/14, Fa0/15
99   Native-Unused                    active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
IIG-Switch#
```

Fig 5: show interfaces trunk — IIG Switch trunk carrying VLANs 10, 20, 30

```
IIG-Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      99
Gig0/2      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gig0/1      10,20,30
Gig0/2      10,20,30

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30
Gig0/2      10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30
Gig0/2      10,20,30

IIG-Switch#
```

Fig 6: show access-lists 150 — ACL 150 with 5 lines and hit counters incrementing during attack

```
Core-Router#show access-lists 150
Extended IP access list 150
    deny tcp 203.0.113.0 0.0.0.255 host 10.0.20.70 eq smtp
    deny icmp 203.0.113.0 0.0.0.255 any echo
    deny icmp 198.51.100.0 0.0.0.255 any echo
    deny icmp 192.0.2.0 0.0.0.255 any echo (1 match(es))
    permit ip any any (1533 match(es))

Core-Router#
```

Fig 7: Ping from Admin PC to Mail Server — successful end-to-end connectivity confirmed

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.20.70

Pinging 10.0.20.70 with 32 bytes of data:

Reply from 10.0.20.70: bytes=32 time=10ms TTL=128
Reply from 10.0.20.70: bytes=32 time=1ms TTL=128
Reply from 10.0.20.70: bytes=32 time<1ms TTL=128
Reply from 10.0.20.70: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.20.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Fig 8: HSRP failover — show standby brief confirming Backup Router promoted to Active

```
Core-Router#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         1    150 P Active   local           10.0.10.13      10.0.10.1
Gig         2    150 P Active   local           10.0.10.45      10.0.10.33
Gig         3    150 P Active   local           10.0.10.61      10.0.10.49
Gig         4    150 P Active   local           10.0.10.19      10.0.10.30
Core-Router#
```

Fig 9: Ping from Zone A PC — blocked by ACL 150, ICMP echo denied

```
PC0
Physical   Config   Desktop   Programming   Attributes
Command Prompt                                                        X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.10.20

Pinging 10.0.10.20 with 32 bytes of data:

Reply from 10.0.10.20: bytes=32 time=10ms TTL=126
Reply from 10.0.10.20: bytes=32 time=11ms TTL=126
Reply from 10.0.10.20: bytes=32 time=10ms TTL=126
Reply from 10.0.10.20: bytes=32 time=1ms TTL=126

Ping statistics for 10.0.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 8ms

C:\>ping 10.0.10.20

Pinging 10.0.10.20 with 32 bytes of data:

Reply from 10.0.10.14: Destination host unreachable.
Reply from 10.0.10.14: Destination host unreachable.
Reply from 10.0.10.14: Destination host unreachable.
Reply from 10.0.10.14: Destination host unreachable.

Ping statistics for 10.0.10.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Fig 10: show ip nat translations — PAT entries confirming inside-to-outside translation

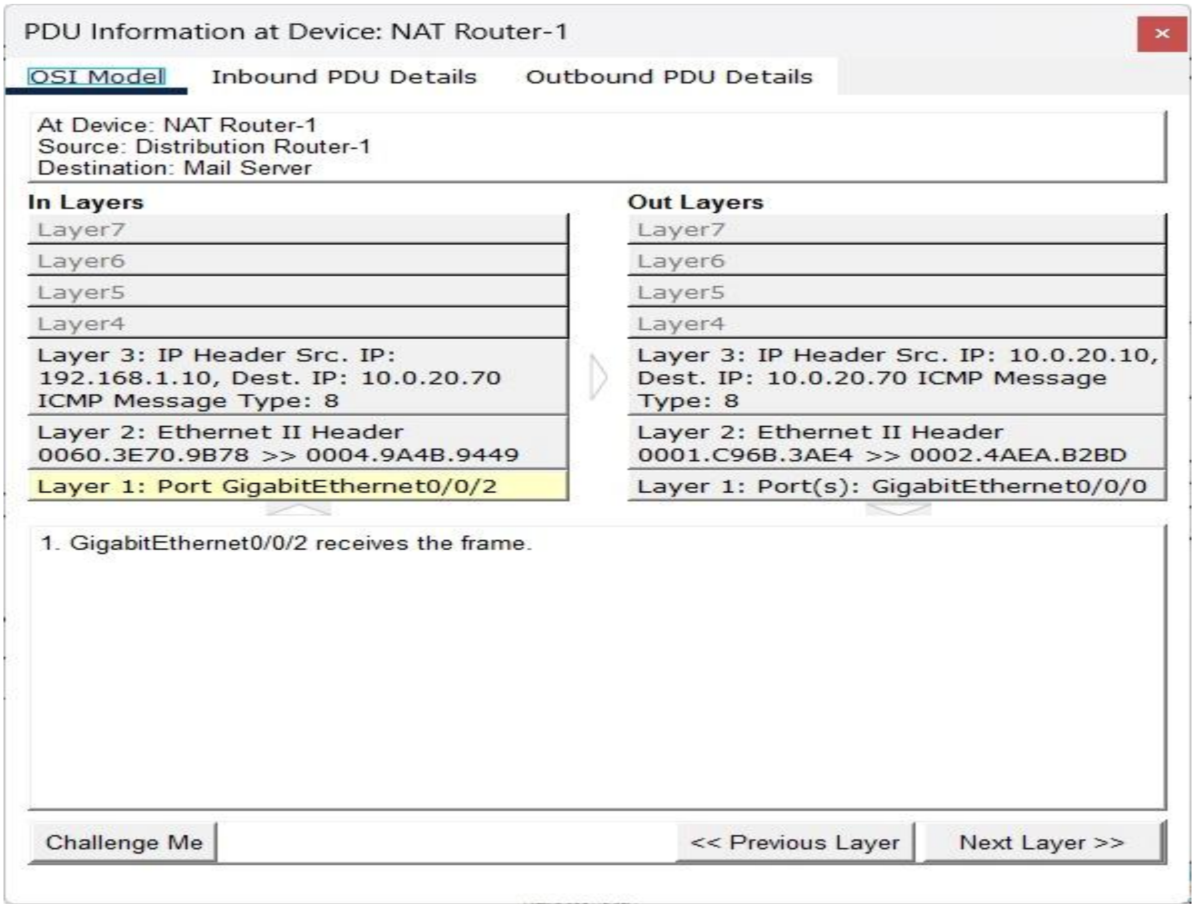


Fig 11: MCU Python script output — 10 flood packets fired with per-packet status

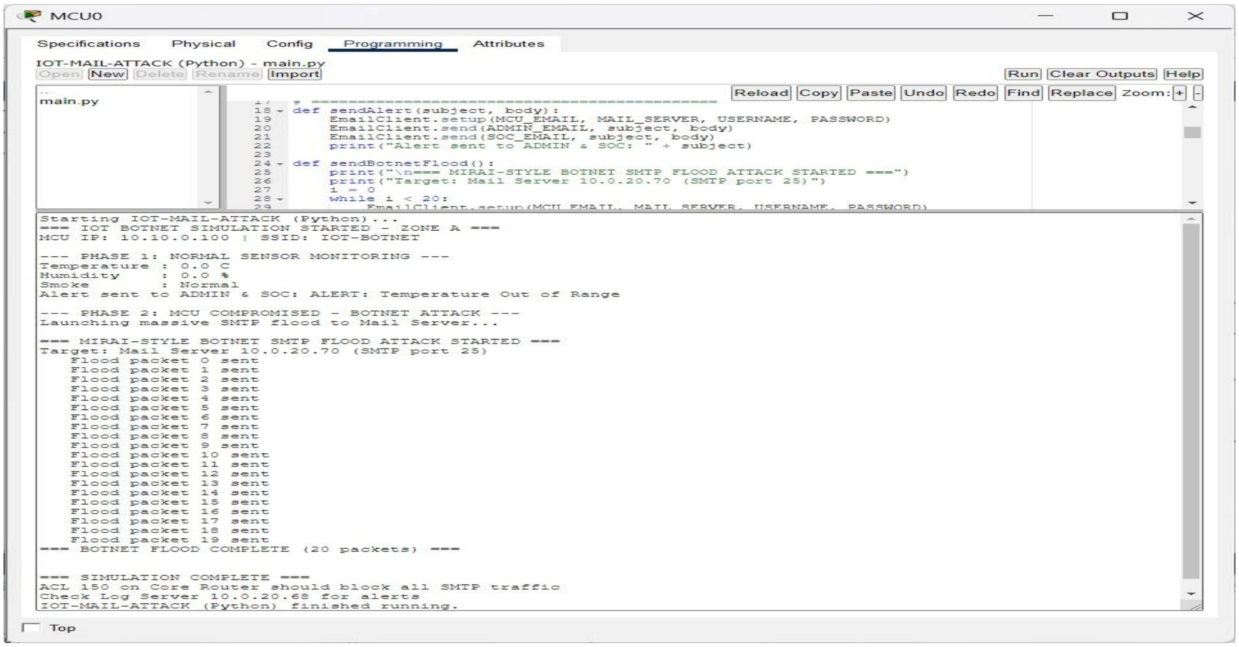


Fig 12: Log Server syslog — ACL deny entries with millisecond-precision timestamps

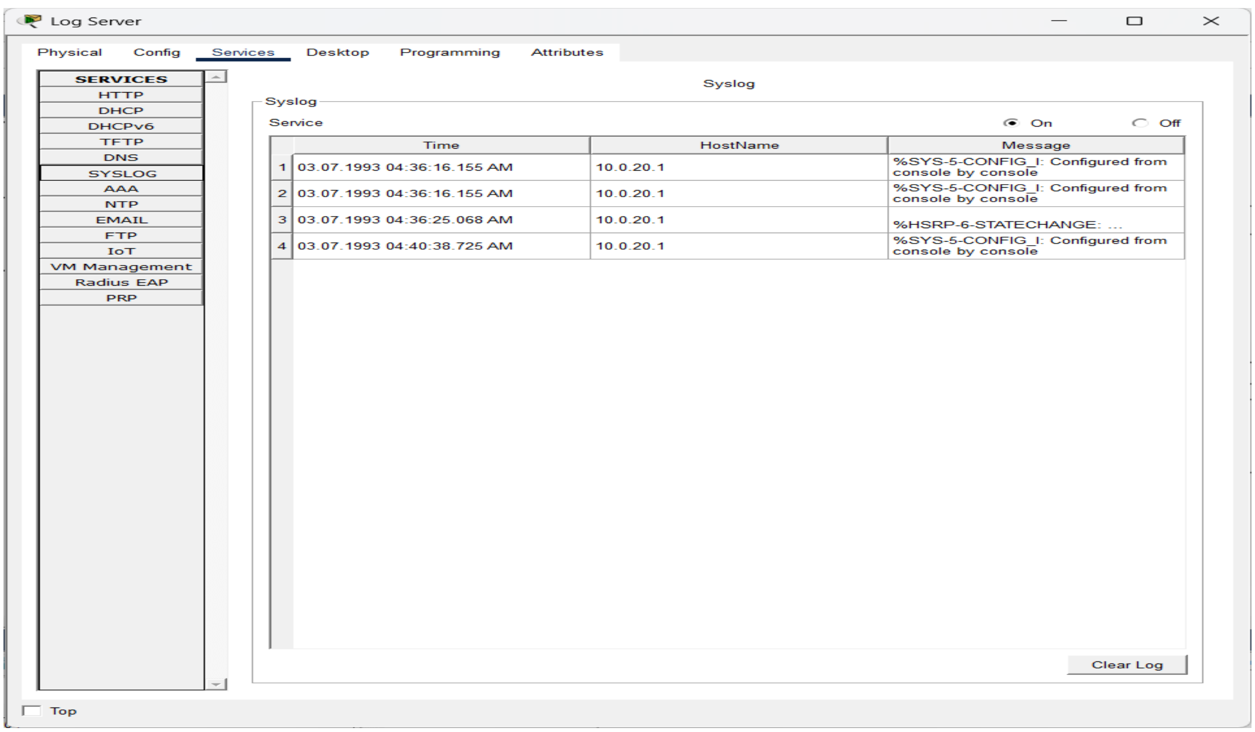
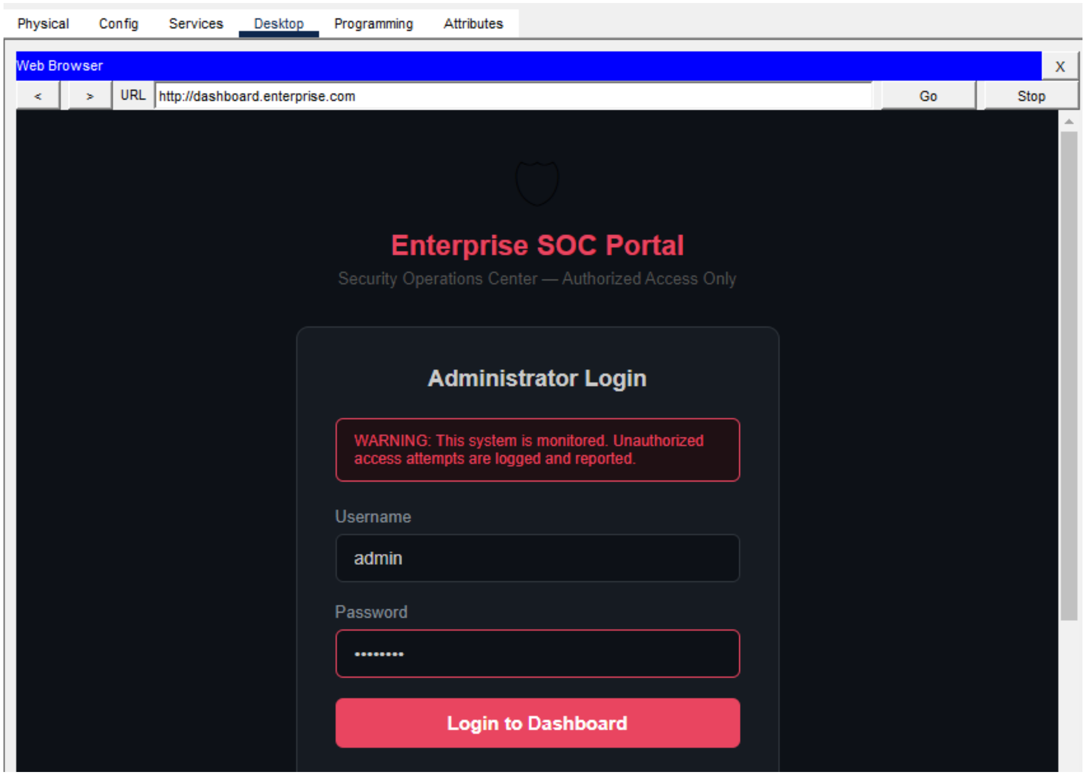
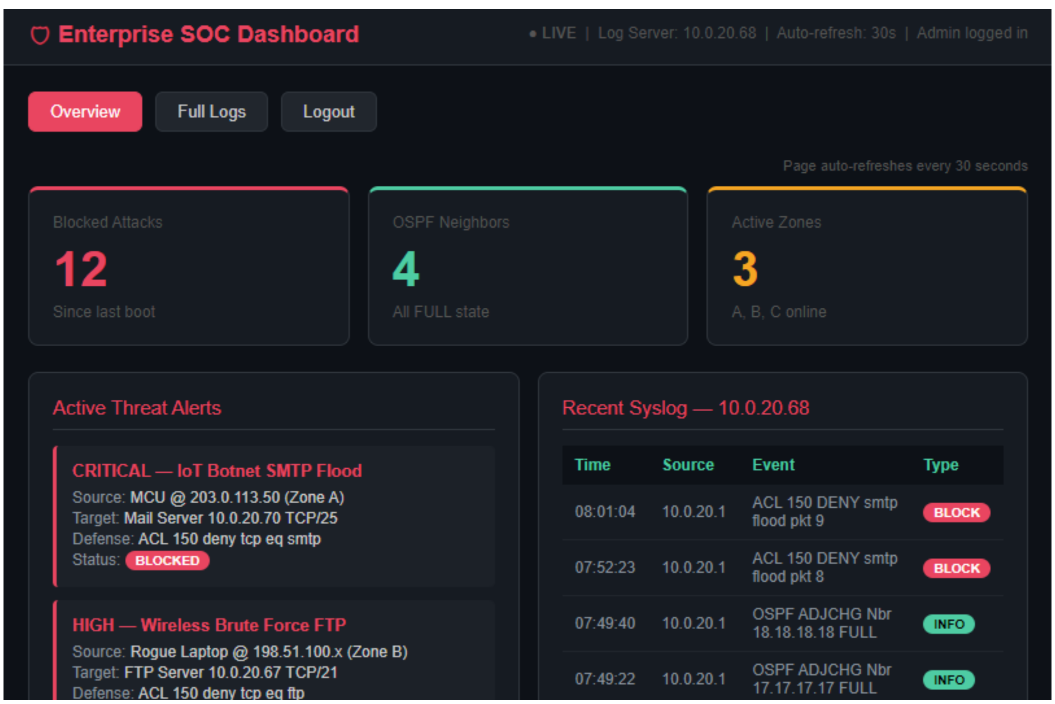


Fig 13: SOC Dashboard login page and main dashboard visible in Admin PC browser





4. Configuration and Implementation

Figure 1 : Configuration on Core Router

Core Router

Physical | Config | CLI | Attributes

IOS

```
==================================================
          IIG NETWORK LAB - CORE ROUTER
          AUTHORIZED ACCESS ONLY!
          Lab 3 - Basic Router Configuration
==================================================


User Access Verification

Password:

Core-Router>en
Core-Router>enable
Password:
Core-Router#show run
Building configuration...

Current configuration : 2313 bytes
!
version 15.4
service timestamps log datetime msec
service timestamps debug datetime msec
service password-encryption
!
hostname Core-Router
!
login block-for 300 attempts 5 within 120
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
ip cef
no ipv6 cef
```

```
!
username admin privilege 15 secret 5 $1$mERr$Pq3/lr0agnISq0fxb9TUZ0
!
!
!
!
!
!
!
!
!
!
no ip domain-lookup
ip domain-name enterprise.com
ip name-server 10.0.20.66
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/0.10
 encapsulation dot1Q 10
 ip address 10.0.10.14 255.255.255.240
 standby 1 ip 10.0.10.1
 standby 1 priority 150
 standby 1 preempt
!
interface GigabitEthernet0/0/0.20
 encapsulation dot1Q 20
 ip address 10.0.10.46 255.255.255.240
 standby 2 ip 10.0.10.33
```

```
 standby 2 priority 150
 standby 2 preempt
!
interface GigabitEthernet0/0/0.30
 encapsulation dot1Q 30
 ip address 10.0.10.62 255.255.255.240
 standby 3 ip 10.0.10.49
 standby 3 priority 150
 standby 3 preempt
!
interface GigabitEthernet0/0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1.40
 encapsulation dot1Q 40
 ip address 10.0.10.17 255.255.255.240
 standby 4 ip 10.0.10.30
 standby 4 priority 150
 standby 4 preempt
!
interface GigabitEthernet0/0/2
 ip address 10.0.20.1 255.255.255.248
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 network 10.0.10.0 0.0.0.15 area 0
 network 10.0.10.16 0.0.0.15 area 0
 network 10.0.10.32 0.0.0.15 area 0
 network 10.0.10.48 0.0.0.15 area 0
 network 10.0.20.0 0.0.0.7 area 0
```

```
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended sl_def_acl
 deny tcp any any eq telnet
 deny tcp any any eq www
 deny tcp any any eq 22
 permit tcp any any eq 22
!
banner motd ^C
============================================
          IIG NETWORK LAB - CORE ROUTER
          AUTHORIZED ACCESS ONLY!
          Lab 3 - Basic Router Configuratior
============================================
^C
!
!
!
!
logging 10.0.20.68
line con 0
 exec-timeout 5 0
 password 7 0822455D0A16
 logging synchronous
 login
!
line aux 0
!
line vty 0 4
 password 7 0822455D0A16
 login local
 transport input ssh
!
!
!
end
```

Core Router: The Core Router is a major component of the enterprise backbone layer, responsible for connecting VLAN segments, internal routing infrastructure, and backbone network resources. It is designed to provide high availability, secure administrative access, dynamic routing, and centralized monitoring for the enterprise environment.

Basic Security and Access Control: The router is secured using an encrypted enable secret and password encryption services to protect locally stored credentials. A privileged administrative account named admin is configured with privilege level 15 for full management access. Login security is strengthened through a blocking mechanism:

- Blocks login for 300 seconds
- After 5 failed attempts within 120 seconds. This helps prevent brute-force password attacks.

Remote management is restricted to SSH-only access on VTY lines using:

- login local
- transport input ssh

Unnecessary DNS lookup is disabled using no ip domain-lookup, while a domain name is configured to support SSH key generation and secure remote login. A Message of the Day (MOTD) banner is also configured to warn unauthorized users.

Inter-VLAN and Sub interface Configuration: Multiple sub interfaces are configured under GigabitEthernet0/0/0 and GigabitEthernet0/0/1 using IEEE 802.1Q encapsulation to support VLAN-based network segmentation. Configured VLAN interfaces include:

- VLAN 10 → 10.0.10.14/28
- VLAN 20 → 10.0.10.46/28
- VLAN 30 → 10.0.10.62/28
- VLAN 40 → 10.0.10.17/28

These subinterfaces enable inter-VLAN routing between user, departmental, and server network segments.

High Availability (HSRP Implementation): HSRP is configured on all VLAN sub interfaces to provide default gateway redundancy. Each VLAN is assigned a virtual IP address, while the Core Router is configured with higher priority 150 and preemption enabled. This ensures that:

- The Core Router remains the preferred active gateway during normal operation
- Automatic failover occurs if the core router becomes unavailable
- Service continuity is maintained through the Backup Router. This removes single points of failure at the gateway level.

Routing Configuration (OSPF): OSPF is implemented as the dynamic routing protocol with Router ID 1.1.1.1. The router operates in Area 0 and advertises all connected VLAN and backbone networks. Advertised networks include:

- 10.0.10.0/28
- 10.0.10.16/28
- 10.0.10.32/28
- 10.0.10.48/28
- 10.0.20.0/29. This allows automatic route discovery, efficient path selection, and fast convergence during link or device failures.

Access Control and Security Policies: A named extended ACL (sl_def_acl) is configured to control management and service traffic. The ACL performs the following actions**:**

- Denies Telnet traffic
- Denies HTTP traffic
- Denies unauthorized TCP port 22 traffic
- Permits SSH traffic

This configuration helps enforce secure remote access policies while blocking insecure management protocols.

Centralized Logging and Monitoring: Syslog is configured to forward logs to the centralized logging server at: 10.0.20.68 This enables:

- Real-time monitoring
- Event tracking
- Security auditing
- Troubleshooting of network incidents. NetFlow export version 9 is also enabled for traffic monitoring and network analysis.

Interface and Backbone Connectivity: Interface GigabitEthernet0/0/2 is configured with IP address 10.0.20.1/29 and serves as the backbone connection between the Core Router and the Core Layer 3 Switch.This link provides high-speed communication between routing layers and supports backbone traffic forwarding across the enterprise network.

Figure 2 : Configuration on Backup Router

Backup Router

Physical Config CLI Attributes

IOS

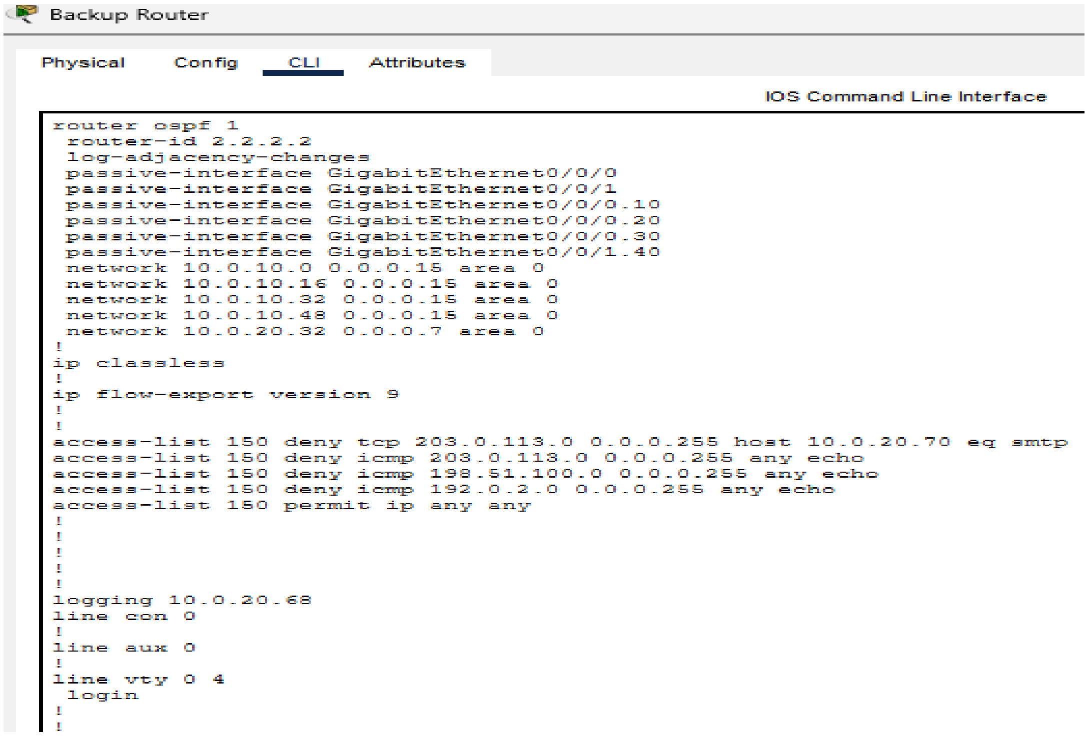
```
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
ip name-server 10.0.20.66
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0/0
 description *** TO IIG SWITCH - HSRP GROUP 1 ***
 no ip address
 ip access-group 150 in
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/0.10
 encapsulation dot1Q 10
 ip address 10.0.10.13 255.255.255.240
 ip access-group 150 in
 standby 1 ip 10.0.10.1
 standby 1 preempt
```

Backup Router

Physical Config CLI Attributes

IOS Commar

```
!
interface GigabitEthernet0/0/0.20
 encapsulation dot1Q 20
 ip address 10.0.10.45 255.255.255.240
 ip access-group 150 in
 standby 2 ip 10.0.10.33
 standby 2 preempt
!
interface GigabitEthernet0/0/0.30
 encapsulation dot1Q 30
 ip address 10.0.10.61 255.255.255.240
 ip access-group 150 in
 standby 3 ip 10.0.10.49
 standby 3 preempt
!
interface GigabitEthernet0/0/1
 description *** TO CACHE SWITCH - HSRP GROUP 2 ***
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1.40
 encapsulation dot1Q 40
 ip address 10.0.10.19 255.255.255.240
 ip access-group 150 in
 standby 4 ip 10.0.10.18
 standby 4 preempt
!
interface GigabitEthernet0/0/2
 description *** TO CORE SWITCH - HSRP GROUP 3 ***
 ip address 10.0.20.3 255.255.255.248
 duplex auto
 speed auto
 standby 3 ip 10.0.20.7
 standby 3 preempt
!
interface Vlan1
 no ip address
 shutdown
```




```
router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
 passive-interface GigabitEthernet0/0/0
 passive-interface GigabitEthernet0/0/1
 passive-interface GigabitEthernet0/0/0.10
 passive-interface GigabitEthernet0/0/0.20
 passive-interface GigabitEthernet0/0/0.30
 passive-interface GigabitEthernet0/0/1.40
 network 10.0.10.0 0.0.0.15 area 0
 network 10.0.10.16 0.0.0.15 area 0
 network 10.0.10.32 0.0.0.15 area 0
 network 10.0.10.48 0.0.0.15 area 0
 network 10.0.20.32 0.0.0.7 area 0
!
ip classless
!
ip flow-export version 9
!
!
access-list 150 deny tcp 203.0.113.0 0.0.0.255 host 10.0.20.70 eq smtp
access-list 150 deny icmp 203.0.113.0 0.0.0.255 any echo
access-list 150 deny icmp 198.51.100.0 0.0.0.255 any echo
access-list 150 deny icmp 192.0.2.0 0.0.0.255 any echo
access-list 150 permit ip any any
!
!
!
!
!
logging 10.0.20.68
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
```

Backup Router: The Backup Router is an essential redundant component of the enterprise backbone network. It ensures continuous network availability in case of failure of the primary routing device. The router supports inter-VLAN routing, dynamic routing, gateway redundancy, and basic security enforcement while operating as a secondary or standby device under normal conditions.

High Availability (HSRP Implementation): HSRP is configured on multiple VLAN subinterfaces to provide default gateway redundancy. The Backup Router participates in HSRP Groups 1, 2, 3, and 4, each associated with different VLAN segments. Virtual IP addresses are configured for each group, allowing end devices to use a single default gateway. The preempt feature is enabled, ensuring that the Backup Router can assume the active role when it has higher priority or when the primary router fails. This configuration eliminates single points of failure and ensures uninterrupted network access.

Inter-VLAN Routing (Subinterface Configuration): The Backup Router uses a router-on-a-stick configuration through GigabitEthernet0/0/0 and GigabitEthernet0/0/1 interfaces. Multiple subinterfaces are created using IEEE 802.1Q encapsulation to support VLAN-based segmentation. Configured VLANs include: VLAN 10 → 10.0.10.13/28, VLAN 20 → 10.0.10.45/28 , VLAN 30 → 10.0.10.61/28 , VLAN 40 → 10.0.10.19/28 . Each subinterface is assigned an IP address within its respective subnet, enabling communication between VLANs when the router is active.

Backbone Connectivity: GigabitEthernet0/0/2 is dedicated to backbone connectivity with the core switch network using the 10.0.20.0/29 subnet. This interface also participates in HSRP Group 3 with virtual IP 10.0.20.7, ensuring redundancy at the core layer.

Routing Configuration (OSPF): OSPF is implemented as the dynamic routing protocol with Router ID 2.2.2.2. The router operates in Area 0 (backbone area) and advertises all VLAN and backbone networks. The following networks are included in OSPF: 10.0.10.0/28 series networks (VLANs), 10.0.20.32/29 backbone network. Passive interfaces are configured on all VLAN and trunk interfaces to prevent unnecessary OSPF neighbor relationships while still allowing route advertisement. This improves security and reduces routing overhead.

Access Control and Security Policies (ACL 150): Extended Access Control List (ACL) 150 is applied inbound on multiple interfaces to enforce basic traffic filtering and security policies. The ACL performs the following functions:

- Blocks SMTP traffic from network 203.0.113.0/24 to host 10.0.20.70
- Denies ICMP echo (ping) requests from the following networks:

    - 203.0.113.0/24
    - 198.51.100.0/24
    - 192.0.2.0/24

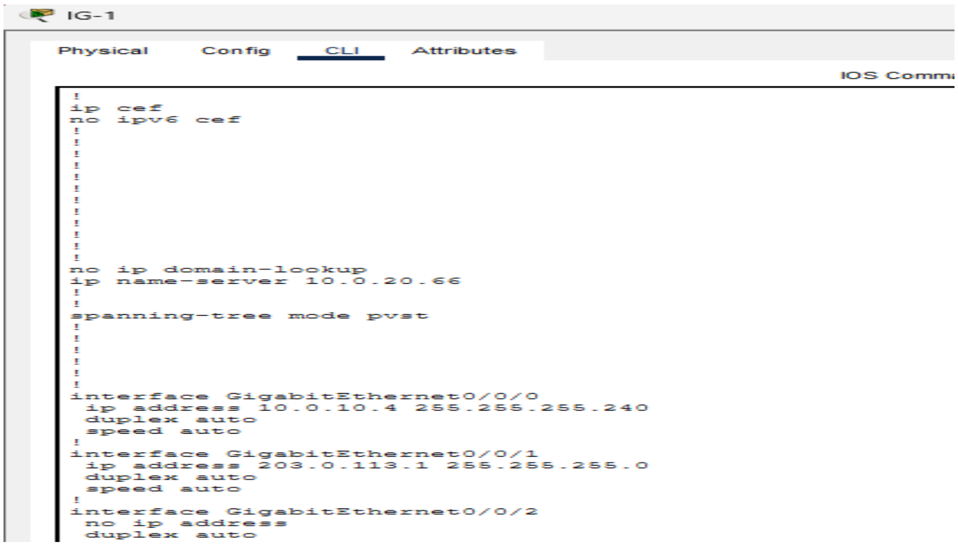
- Permits all other IP traffic

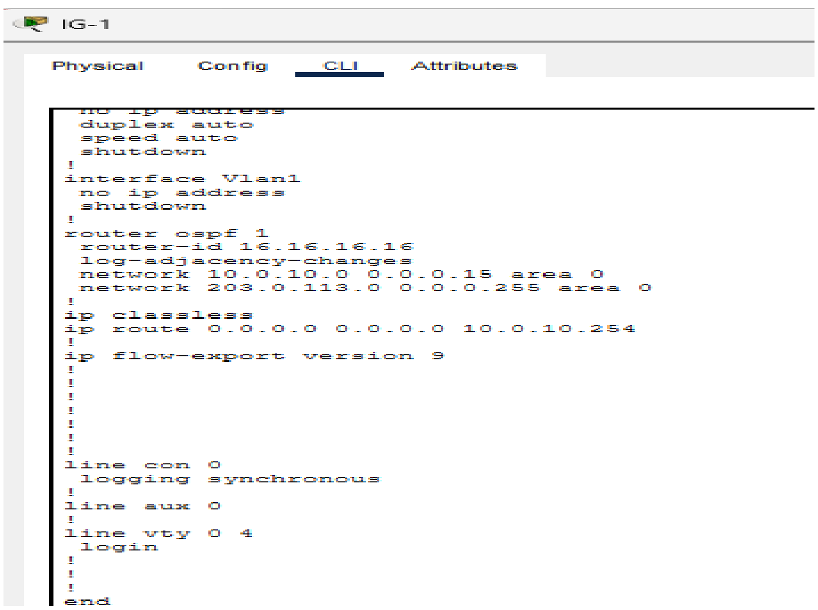
Network Services and Monitoring:

- A DNS server (10.0.20.66) is configured for name resolution within the network.
- Syslog logging is enabled to forward logs to 10.0.20.68 for centralized monitoring and troubleshooting.
- Cisco Express Forwarding (CEF) is enabled to optimize packet forwarding performance.
- Spanning Tree Protocol (PVST) is enabled for loop prevention in Layer 2 switching environments.

Interface Configuration Summary: The router uses trunk-based subinterfaces on GigabitEthernet0/0/0 and GigabitEthernet0/0/1 to support VLAN segmentation. All interfaces are integrated with HSRP and ACL policies, ensuring both redundancy and security.

Figure 3 : Configuration on IG Router





Internet Gateway (IG) Router: The IG Router (Internet Gateway Router) acts as the boundary device between the internal enterprise network and external/public networks. It is responsible for routing traffic between private IP networks and the Internet while participating in internal dynamic routing to ensure connectivity across the enterprise infrastructure.

Interface Configuration: The router is configured with two active interfaces

- **GigabitEthernet0/0/0 (Internal Network Interface)** IP address: 10.0.10.4/28 This interface connects the router to the internal enterprise network and participates in OSPF routing for internal communication.
- **GigabitEthernet0/0/1(External Network Interface)** IP address: 203.0.113.1/24 This interface connects to the external/public network and acts as the Internet-facing interface of the router.

- **GigabitEthernet0/0/2** is administratively shutdown and not used in this configuration.

Routing Configuration (OSPF): OSPF is configured as the dynamic routing protocol with Router ID 16.16.16.16. The router operates in Area 0 and advertises both internal and external networks.

- 10.0.10.0/28 (Internal enterprise network)
- 203.0.113.0/24 (External network segment). This enables efficient route exchange between internal routers and ensures network reachability across different segments.

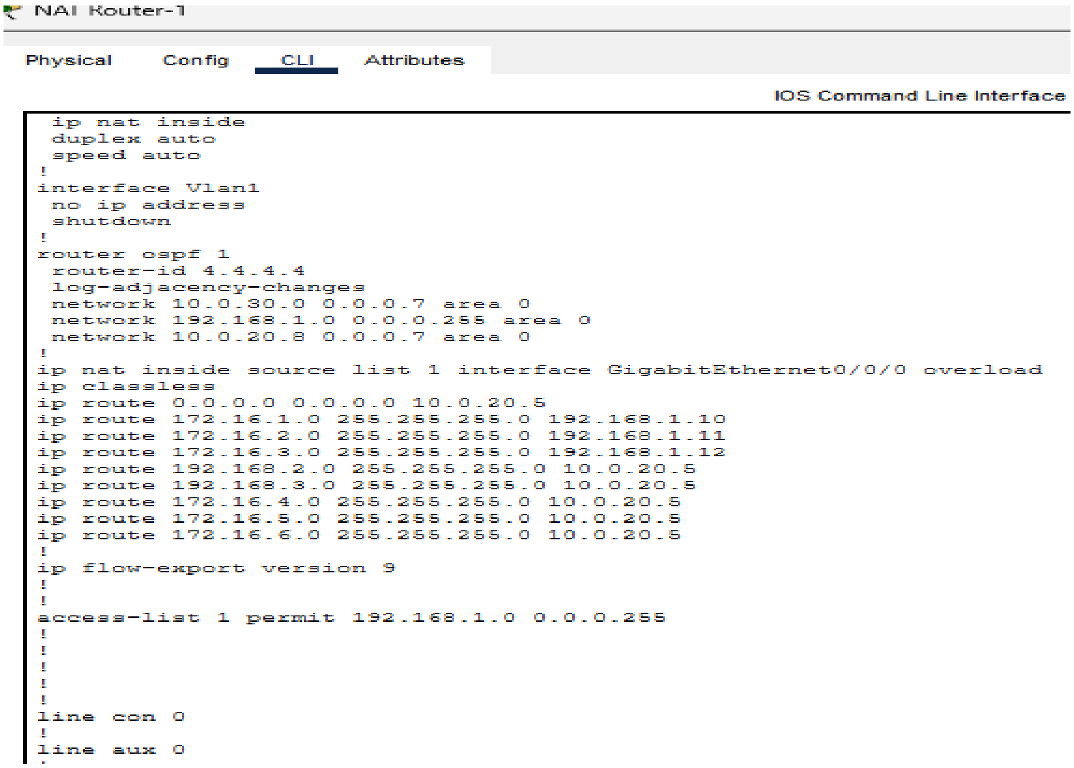
Default Route Configuration: A static default route is configured.

- 0.0.0.0/0 → 10.0.10.254 . This ensures that all traffic destined for unknown networks is forwarded toward the internal next-hop router, which handles further routing toward external networks.

Network Services and Basic Settings:

- DNS server is configured as 10.0.20.66 for name resolution services.
- no ip domain-lookup is enabled to prevent delays caused by incorrect CLI inputs.
- Cisco Express Forwarding (CEF) is enabled to improve packet forwarding efficiency.
- Spanning Tree Protocol (PVST) is enabled for loop prevention in connected Layer 2 environments.

Figure 4 : Configuration on NAT Router




```
 ip nat inside
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
 network 10.0.30.0 0.0.0.7 area 0
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.20.8 0.0.0.7 area 0
!
ip nat inside source list 1 interface GigabitEthernet0/0/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 10.0.20.5
ip route 172.16.1.0 255.255.255.0 192.168.1.10
ip route 172.16.2.0 255.255.255.0 192.168.1.11
ip route 172.16.3.0 255.255.255.0 192.168.1.12
ip route 192.168.2.0 255.255.255.0 10.0.20.5
ip route 192.168.3.0 255.255.255.0 10.0.20.5
ip route 172.16.4.0 255.255.255.0 10.0.20.5
ip route 172.16.5.0 255.255.255.0 10.0.20.5
ip route 172.16.6.0 255.255.255.0 10.0.20.5
!
ip flow-export version 9
!
!
access-list 1 permit 192.168.1.0 0.0.0.255
!
!
!
!
!
line con 0
!
line aux 0
```

NAT Router: The NAT Router (Network Address Translation Router) is a key boundary device in the enterprise network responsible for translating private IP addresses into a public-facing address. It enables multiple internal devices to access external networks using a single routable IP address while maintaining security and address conservation

Interface Configuration: The NAT Router is configured with three active interfaces, each assigned a specific role:

- **GigabitEthernet0/0/0 (Outside Interface)** IP address: 10.0.20.10/29
  This interface connects to the core switch and is defined as the **NAT outside interface**, representing the external-facing side of the network.
- **GigabitEthernet0/0/2 (Inside Interface)** IP address: 192.168.1.1/24
  This interface connects to the distribution layer switches and is defined as the **NAT inside interface**, representing the internal private network.
- **GigabitEthernet0/0/1** IP address: 10.0.30.1/29
  This interface is used for additional internal routing connectivity and participates in the internal network infrastructure.

NAT Configuration: Network Address Translation (NAT) is configured using an access control list (ACL 1) to identify the internal private network

- Allowed network: 192.168.1.0/24

The following NAT rule is applied:

- ip nat inside source list 1 interface GigabitEthernet0/0/0 overload

This configuration enables PAT (Port Address Translation), allowing multiple internal devices to share a single external IP address. This ensures efficient IPv4 address usage and controlled external access.

Routing Configuration (OSPF and Static Routes): OSPF is configured with Router ID 4.4.4.4 and operates in Area 0. It advertises the following networks : 10.0.30.0/29 , 192.168.1.0/24, 10.0.20.8/29 . In addition to OSPF, multiple static routes are configured to ensure reachability to remote internal networks and distribution segments. A default route is also configured:

- 0.0.0.0/0 → 10.0.20.5 This ensures that unknown traffic is forwarded toward the upstream core network.

Security Configuration:

- Enable secret password is configured and encrypted for privileged access security.
- service password-encryption is enabled to protect configuration passwords.
- Access Control List 1 is used to define the internal network eligible for NAT translation.
- no ip domain-lookup is enabled to prevent delays caused by incorrect command inputs.

Network Services:

- DNS server is configured as 10.0.20.66 for name resolution services.
- Cisco Express Forwarding (CEF) is not explicitly enabled in this configuration, but basic IP routing is active.
- Spanning Tree Protocol (PVST) is enabled for Layer 2 loop prevention in connected switch environments.

Figure 5 : Configuration on Distribution Router

Distribution Router-1

Physical Config CLI Attributes

```
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
ip name-server 10.0.20.66
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
 ip address 192.168.1.10 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 ip address 172.16.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
```

```
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 7.7.7.7
 log-adjacency-changes
 network 192.168.1.0 0.0.0.255 area 0
 network 172.16.1.0 0.0.0.255 area 0
!
ip classless
ip route 10.0.20.0 255.255.255.248 192.168.1.1
ip route 10.0.20.4 255.255.255.252 192.168.1.1
ip route 10.0.20.8 255.255.255.252 192.168.1.1
ip route 10.0.20.12 255.255.255.252 192.168.1.1
ip route 10.0.20.64 255.255.255.240 192.168.1.1
ip route 10.0.10.0 255.255.255.240 192.168.1.1
ip route 10.0.10.16 255.255.255.240 192.168.1.1
ip route 192.168.2.0 255.255.255.0 192.168.1.1
ip route 192.168.3.0 255.255.255.0 192.168.1.1
ip route 172.16.0.0 255.255.0.0 192.168.1.1
!
ip flow-export version 9
```

```
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
```

Distribution Router: The Distribution Router is an important intermediate routing device within the enterprise network. It operates between the access layer and the core/NAT layer, aggregating traffic from internal sub networks and forwarding it toward upstream routers. The router supports internal routing, subnet connectivity, and controlled communication between local departmental networks and the backbone infrastructure.

Interface Configuration: The router is configured with two active interfaces

- **GigabitEthernet0/0/0 (Uplink Interface)** IP address: 192.168.1.10/24 This interface connects the Distribution Router to the NAT Router or upstream internal routing segment. It serves as the primary uplink toward the core network.
- **GigabitEthernet0/0/1 (Local LAN Interface)** IP address: 172.16.1.1/24 This interface connects to the local departmental or access network and acts as the default gateway for hosts within the 172.16.1.0/24 subnet.
- **GigabitEthernet0/0/2** is administratively shut down and not currently in use.

Routing Configuration (OSPF): OSPF is configured as the dynamic routing protocol with Router ID 7.7.7.7. The router operates in Area 0 and advertises its directly connected networks

- 192.168.1.0/24
- 172.16.1.0/24. This allows automatic route exchange with neighboring routers and ensures internal network reachability.

Static Route Configuration: Multiple static routes are configured with next-hop address 192.168.1.1, which points toward the upstream NAT Router. These routes provide connectivity to remote enterprise networks, server segments, and backbone subnets. Configured destination networks include

- 10.0.20.0/29
- 10.0.20.4/30
- 10.0.20.8/30
- 10.0.20.12/30
- 10.0.20.64/28
- 10.0.10.0/28
- 10.0.10.16/28
- 192.168.2.0/24
- 192.168.3.0/24
- 172.16.0.0/16

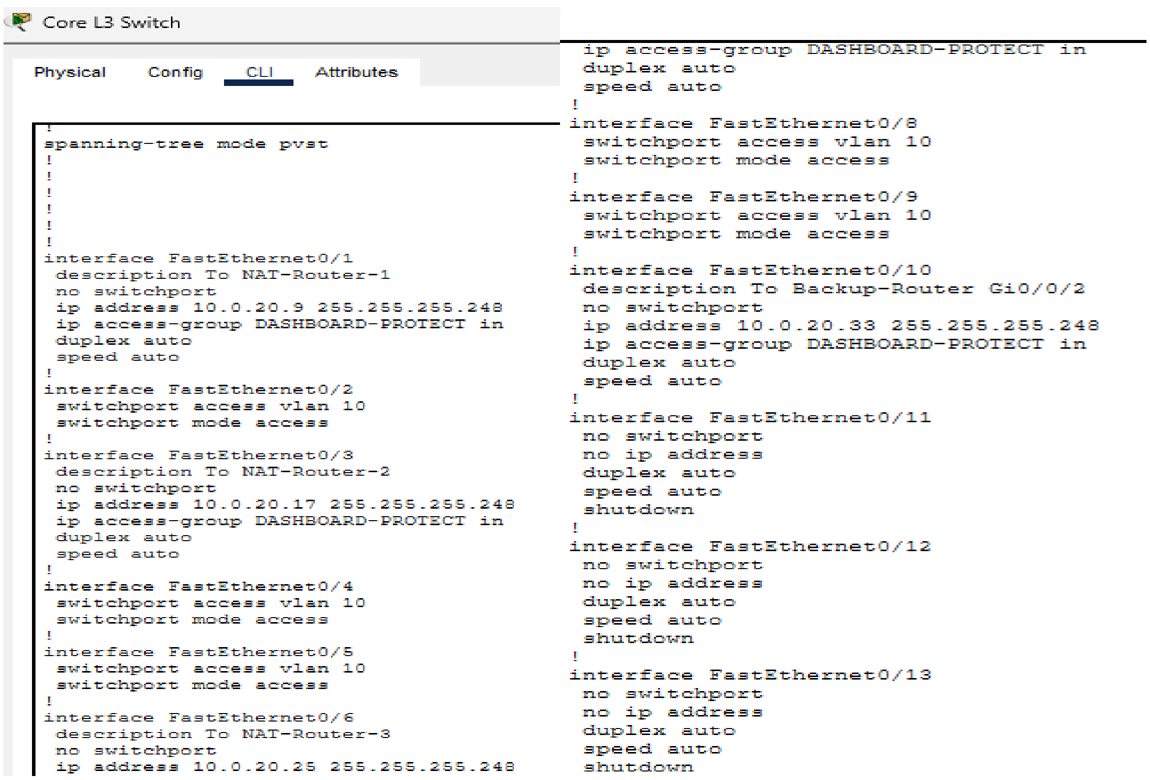
This hybrid routing design combines OSPF dynamic routing with static route control for predictable traffic forwarding.

Network Services and System Settings:

- DNS server is configured as 10.0.20.66 for hostname resolution.
- Cisco Express Forwarding (CEF) is enabled to improve routing performance.
- Spanning Tree Protocol (PVST) is enabled for loop prevention in connected switching environments.
- IP classless routing is enabled to support modern subnet-based forwarding.

Security and Remote Access: Basic remote management is configured through VTY lines with login enabled. Console and auxiliary line access are also available for local administration.

Figure 6 : Configuration on Core L3 Switch



```
interface FastEthernet0/16
 no switchport
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet0/17
 no switchport
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 description === GATEWAY FOR SWITCH 6 BACKBONE
 no ip address
!
interface Vlan10
 mac-address 0001.43a7.3101
 ip address 10.0.20.65 255.255.255.240
 ip helper-address 10.0.20.66
```

```
 ip helper-address 10.0.20.66
!
router ospf 1
 router-id 3.3.3.3
 log-adjacency-changes
 network 10.0.20.64 0.0.0.15 area 0
 network 10.0.20.0 0.0.0.7 area 0
 network 10.0.20.8 0.0.0.7 area 0
 network 10.0.20.16 0.0.0.7 area 0
 network 10.0.20.24 0.0.0.7 area 0
 network 10.0.20.32 0.0.0.7 area 0
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended DASHBOARD-PROTECT
 permit tcp host 10.0.20.69 host 10.0.20.67 eq www
 deny tcp any host 10.0.20.67 eq www
 permit ip any any
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
!
end
```

Core Layer 3 Switch: The Core Layer 3 Switch is the central backbone switching and routing device of the enterprise network. It combines Layer 2 switching with Layer 3 routing functions to interconnect NAT routers, the primary core router, the backup router, and internal VLAN segments. The switch is designed to provide high-speed packet forwarding, centralized connectivity, secure traffic control, and scalable backbone performance.

Layer 3 Routing Configuration: IP routing is enabled on the switch, allowing it to route traffic between different subnets and routed interfaces. This converts the device from a normal Layer 2 switch into a multilayer switch capable of performing internal backbone routing. This configuration provides:

- Communication between multiple connected networks
- Faster internal traffic forwarding
- Reduced load on external routers
- Efficient backbone routing performance

Routed Interface Configuration: Several FastEthernet interfaces are configured as Layer 3 routed ports using the no switchport command. These ports are used as direct point-to-point links to routers and NAT devices. Configured routed interfaces include:

- FastEthernet0/1 → NAT-Router-1
  IP address: 10.0.20.9/29
- FastEthernet0/3 → NAT-Router-2
  IP address: 10.0.20.17/29
- FastEthernet0/6 → NAT-Router-3
  IP address: 10.0.20.25/29
- FastEthernet0/7 → Core Router
  IP address: 10.0.20.2/29
- FastEthernet0/10 → Backup Router
  IP address: 10.0.20.33/29

These interfaces allow direct routed communication between the Core Layer 3 Switch and connected network devices.

VLAN and Access Port Configuration: Several switch ports are configured as access ports in VLAN 10 to support local host or downstream connections. Configured access ports include:

- FastEthernet0/2
- FastEthernet0/4
- FastEthernet0/5
- FastEthernet0/8
- FastEthernet0/9

This segmentation allows organized network access and simplified traffic management.

Switch Virtual Interface (SVI): A Switch Virtual Interface is configured for VLAN 10 with the following IP address

- VLAN 10 → 10.0.20.65/28

This interface acts as the default gateway for devices connected to VLAN 10 and enables routing between VLAN 10 and other backbone networks.

A helper address is also configured:

- ip helper-address 10.0.20.66. This forwards DHCP broadcast requests to the DHCP server, allowing centralized IP address assignment for VLAN users.

Routing Configuration (OSPF): OSPF is configured as the dynamic routing protocol with Router ID 3.3.3.3. The switch operates in Area 0 and advertises all major backbone and VLAN networks. Advertised networks include:

- 10.0.20.64/28
- 10.0.20.0/29
- 10.0.20.8/29
- 10.0.20.16/29
- 10.0.20.24/29
- 10.0.20.32/29

This allows automatic route exchange with connected routers and supports efficient traffic forwarding throughout the enterprise network.

Access Control and Security Policies: A named extended ACL called DASHBOARD-PROTECT is applied inbound on multiple routed interfaces. The ACL performs the following actions

- Permits HTTP access from host 10.0.20.69 to host 10.0.20.67
- Denies HTTP access from all other sources to host 10.0.20.67
- Permits all remaining IP traffic

This configuration protects the internal web dashboard server by allowing access only from an authorized management host while blocking unauthorized users.

System Features and Monitoring: The switch includes several operational features

- PVST is enabled for loop prevention in switching environments
- IP classless routing is enabled
- NetFlow export version 9 is configured for traffic monitoring and analysis
- Unused routed ports are administratively shut down to improve security and reduce unnecessary exposure

Administrative Access: Console, auxiliary, and VTY lines are configured for management access. Remote administration is available through VTY login configuration.

Figure 7 : Configuration on IG Switch

IG switch

Physical Config CLI Attributes

```
hostname IIG-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
```

```
interface FastEthernet0/11
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/13
 switchport access vlan 30
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/14
 switchport access vlan 30
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/15
 switchport access vlan 30
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
```

```
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
!
interface GigabitEthernet0/2
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
!
interface Vlan1
 no ip address
 shutdown
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
!
```

IG Switch: The IG Switch is an access-layer switch responsible for connecting end-user devices to the enterprise network and forwarding traffic toward the gateway routers through trunk uplinks. It provides VLAN-based segmentation, efficient Layer 2 switching, and organized user distribution across multiple departments or network zones.

Switching and VLAN Configuration: The switch is configured in Layer 2 switching mode and uses VLAN segmentation to separate user traffic into different broadcast domains. Multiple access ports are assigned to specific VLANs based on network requirements. Configured VLAN assignments include:

- **VLAN 10** FastEthernet0/1 – FastEthernet0/6 FastEthernet0/11
- **VLAN 20** FastEthernet0/7 – FastEthernet0/10
- **VLAN 30** FastEthernet0/12 – FastEthernet0/15

This VLAN structure improves network performance, security, and traffic management by isolating users into separate logical groups.

Access Port Configuration: FastEthernet interfaces connected to end devices are configured as access ports using switchport mode access. Each port is assigned to a designated VLAN depending on user location or department. All access ports are configured with: Spanning Tree PortFast. PortFast allows end devices such as PCs and printers to begin communication immediately without waiting for normal Spanning Tree convergence delays. This improves startup connectivity for client devices.

Trunk Link Configuration: Two GigabitEthernet interfaces are configured as trunk ports:

- **GigabitEthernet0/1**
- **GigabitEthernet0/2**

Both trunk links use:

- Native VLAN: 99
- Allowed VLANs: 10, 20, 30

These trunk ports are used to carry traffic from multiple VLANs simultaneously toward connected routers or upstream switches. This ensures:

- Centralized uplink connectivity

- VLAN traffic transport over a single physical link
- Support for inter-VLAN routing through external Layer 3 devices

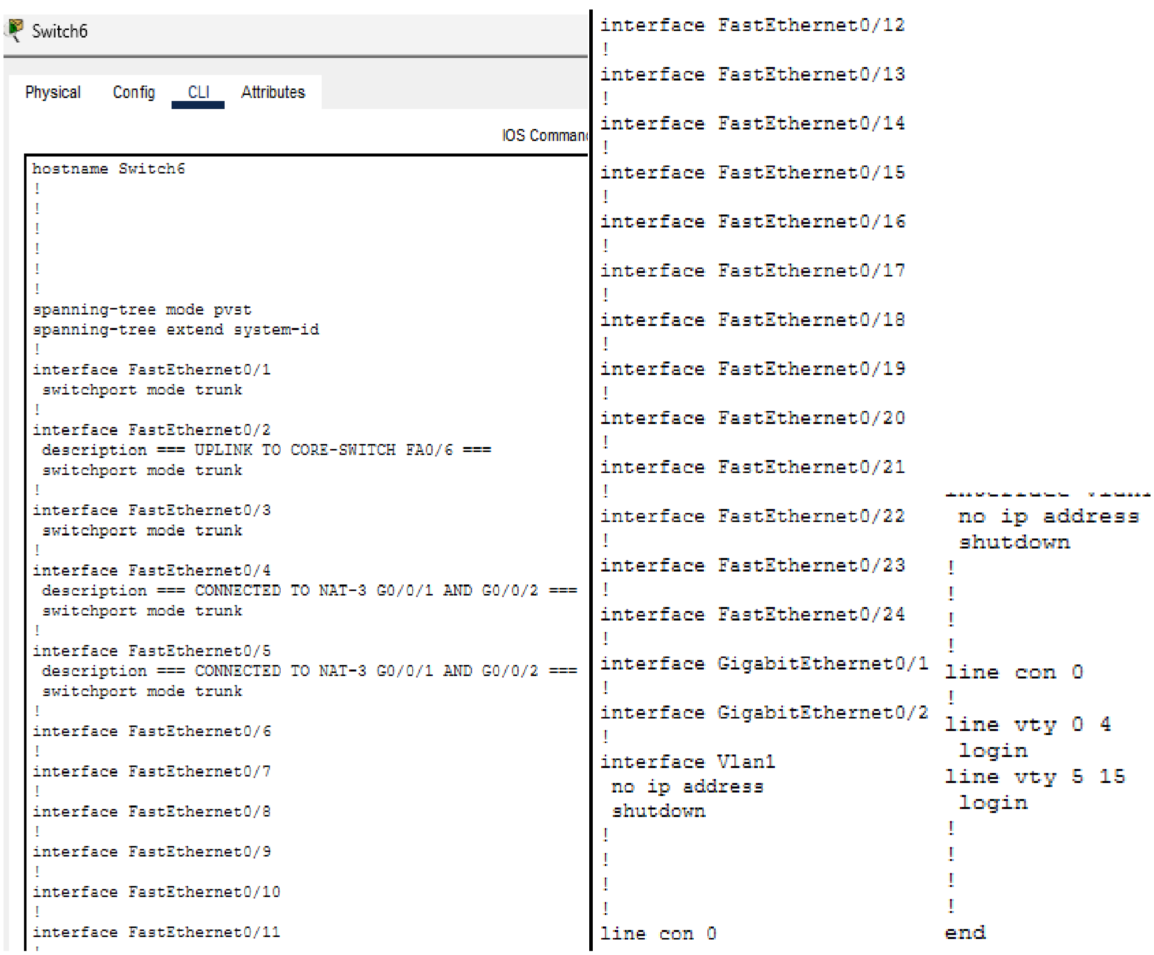
Spanning Tree Configuration: The switch operates using Per-VLAN Spanning Tree Protocol (PVST) with extended system ID enabled. This provides:

- Loop prevention in redundant switching environments
- Separate spanning tree instances for each VLAN
- Improved network stability and failover control

Management Configuration:

- VLAN 1 interface is administratively shut down and has no IP address assigned.
- Console and VTY lines are enabled for local and remote administrative login. Disabling unused management VLAN interfaces reduces unnecessary exposure and improves security.

Figure 8 : Configuration on Switch 6




```
hostname Switch6
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 description === UPLINK TO CORE-SWITCH FA0/6 ===
 switchport mode trunk
!
interface FastEthernet0/3
 switchport mode trunk
!
interface FastEthernet0/4
 description === CONNECTED TO NAT-3 G0/0/1 AND G0/0/2 ===
 switchport mode trunk
!
interface FastEthernet0/5
 description === CONNECTED TO NAT-3 G0/0/1 AND G0/0/2 ===
 switchport mode trunk
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
 shutdown
!
!
!
!
line con 0
```

```
 no ip address
 shutdown
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
!
!
!
!
end
```

Switch 6: Switch 6 is an access/distribution layer Layer 2 switch connected directly to the Core Layer 3 Switch. It functions as an aggregation switch for trunk-based connectivity, forwarding VLAN traffic between the core backbone and connected NAT devices or downstream network segments. The switch is designed to extend VLAN communication and provide scalable switching infrastructure within the enterprise network.

Uplink Connectivity: The switch is connected to the Core Layer 3 Switch through an uplink trunk interface

- **FastEthernet0/2**
  Description: Uplink to Core Switch

This trunk connection allows multiple VLANs to pass between Switch 6 and the Core Layer 3 Switch over a single physical link. It enables centralized routing and backbone communication through the core infrastructure.

Trunk Port Configuration: Several interfaces are configured in trunk mode to carry traffic from multiple VLANs simultaneously. Configured trunk ports include:

- FastEthernet0/1
- FastEthernet0/2
- FastEthernet0/3
- FastEthernet0/4
- FastEthernet0/5. These trunk interfaces are used for inter-switch links and router/switch connectivity, allowing VLAN traffic to be transported efficiently across the network.

NAT Device Connectivity: Interfaces FastEthernet0/4 and FastEthernet0/5 are connected to NAT-3 interfaces, as indicated in the port descriptions. This configuration suggests that Switch 6 is being used to extend connectivity between the Core Layer 3 Switch and the NAT Router environment, allowing VLAN or multi-network traffic exchange through trunk links.

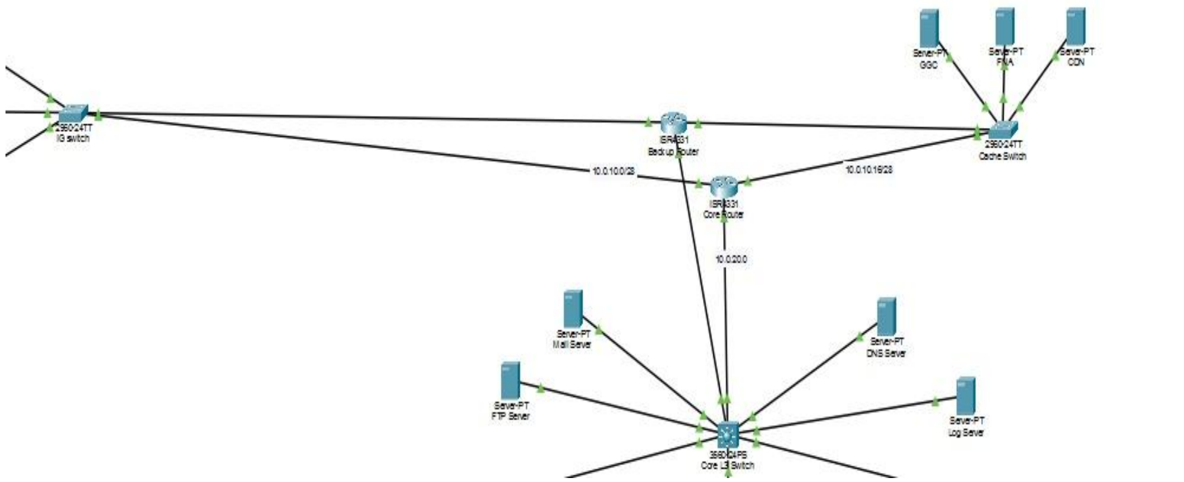
Spanning Tree Configuration: The switch operates using Per-VLAN Spanning Tree Protocol (PVST) with extended system ID enabled. This provides:

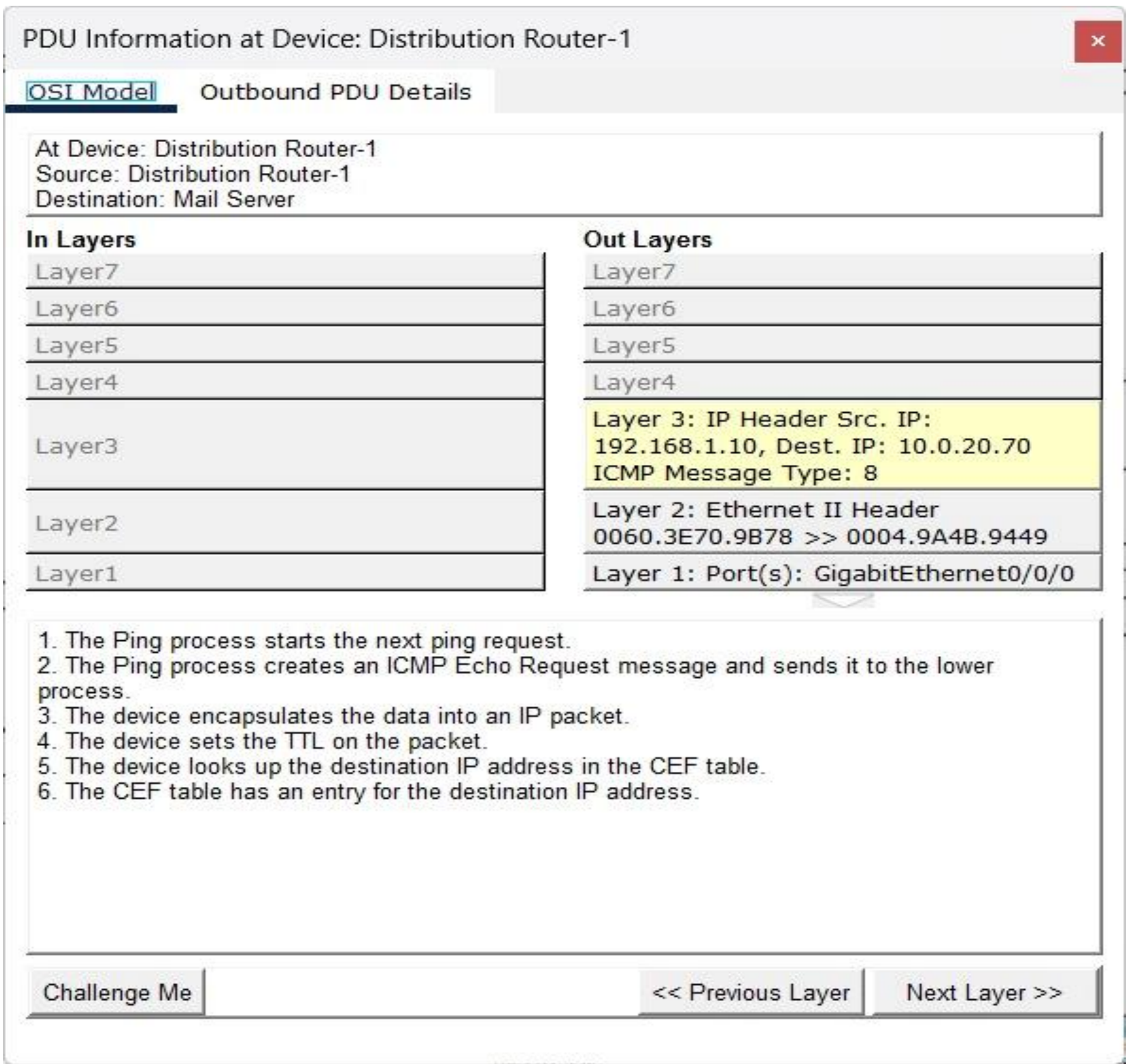
- Prevention of Layer 2 switching loops
- Separate spanning tree instances for each VLAN
- Stable failover behavior in redundant switched environments

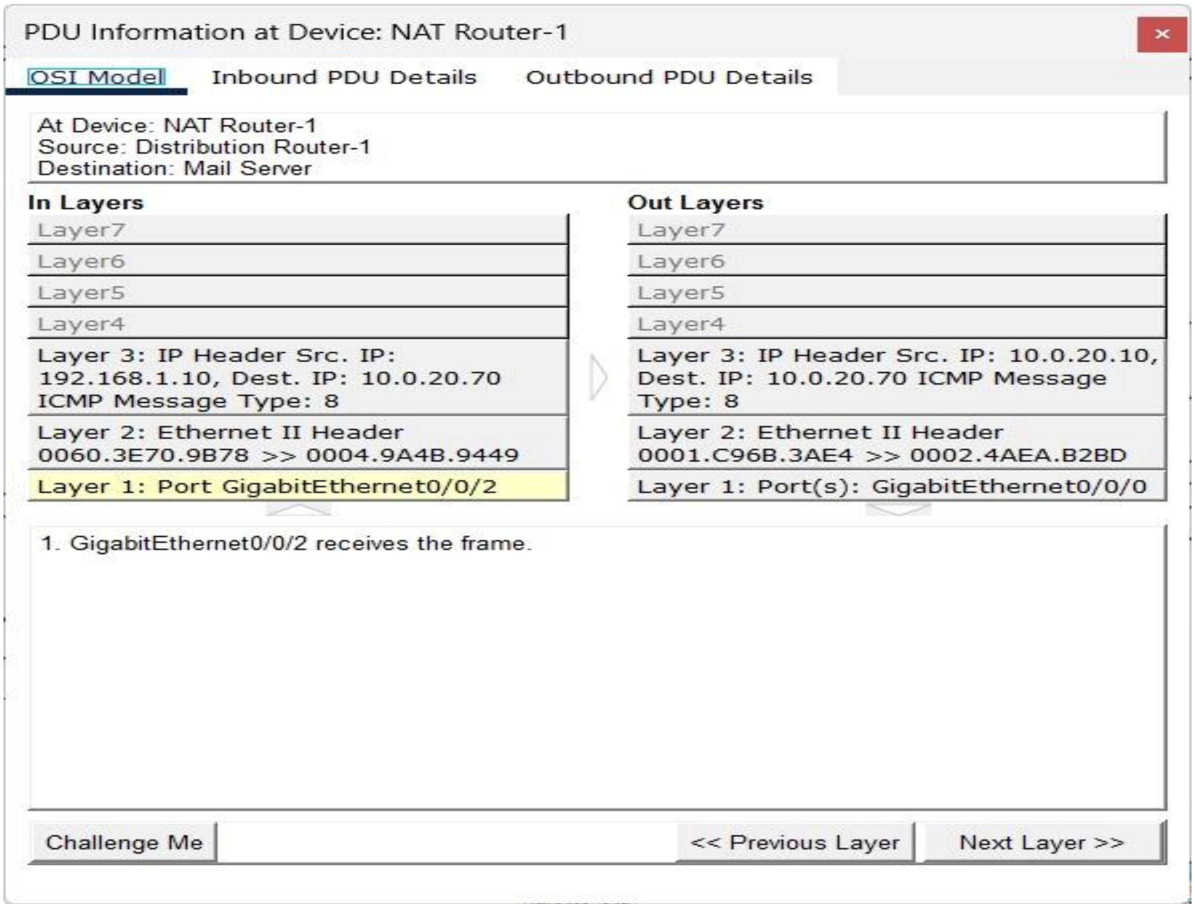
Management and Security:

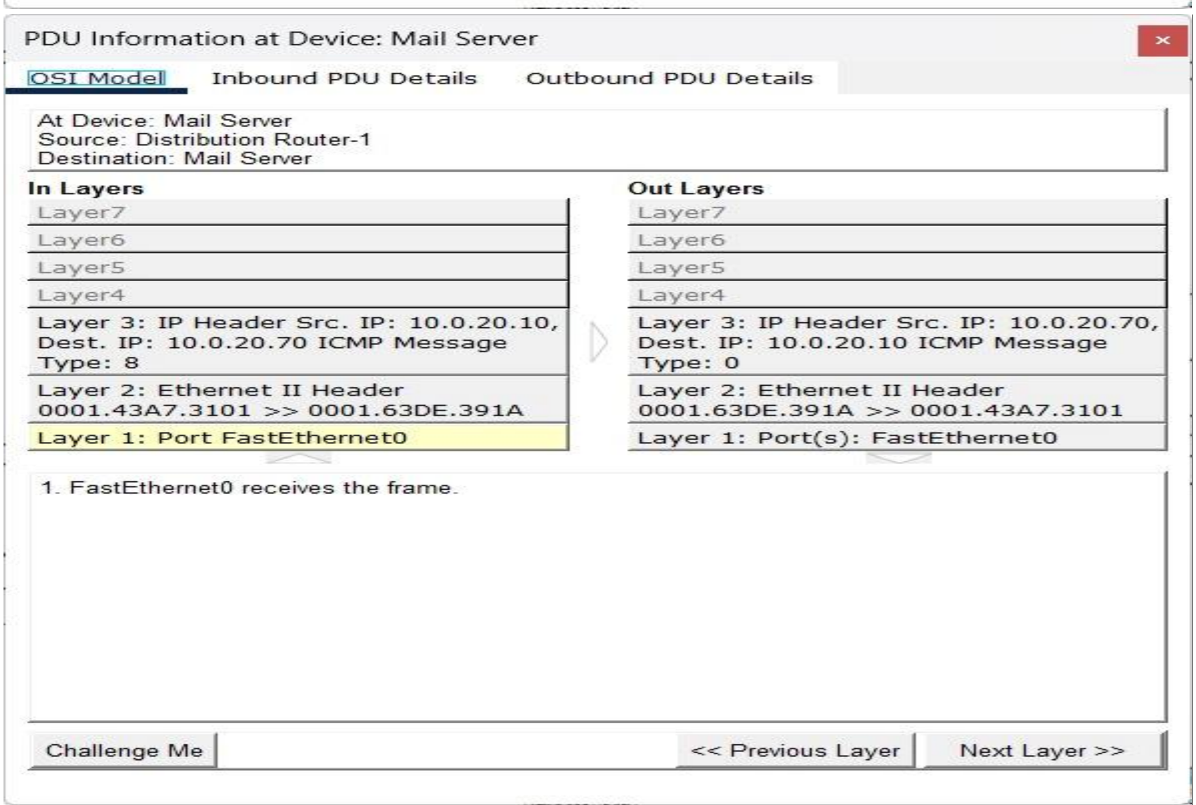
- VLAN 1 interface is administratively shut down and has no IP address assigned.
- Console and VTY lines are enabled for administrative access.
- Unused interfaces remain available for future expansion. Disabling the default VLAN interface reduces unnecessary management exposure and supports better security practice.

Figure 9 : NAT Translation Process



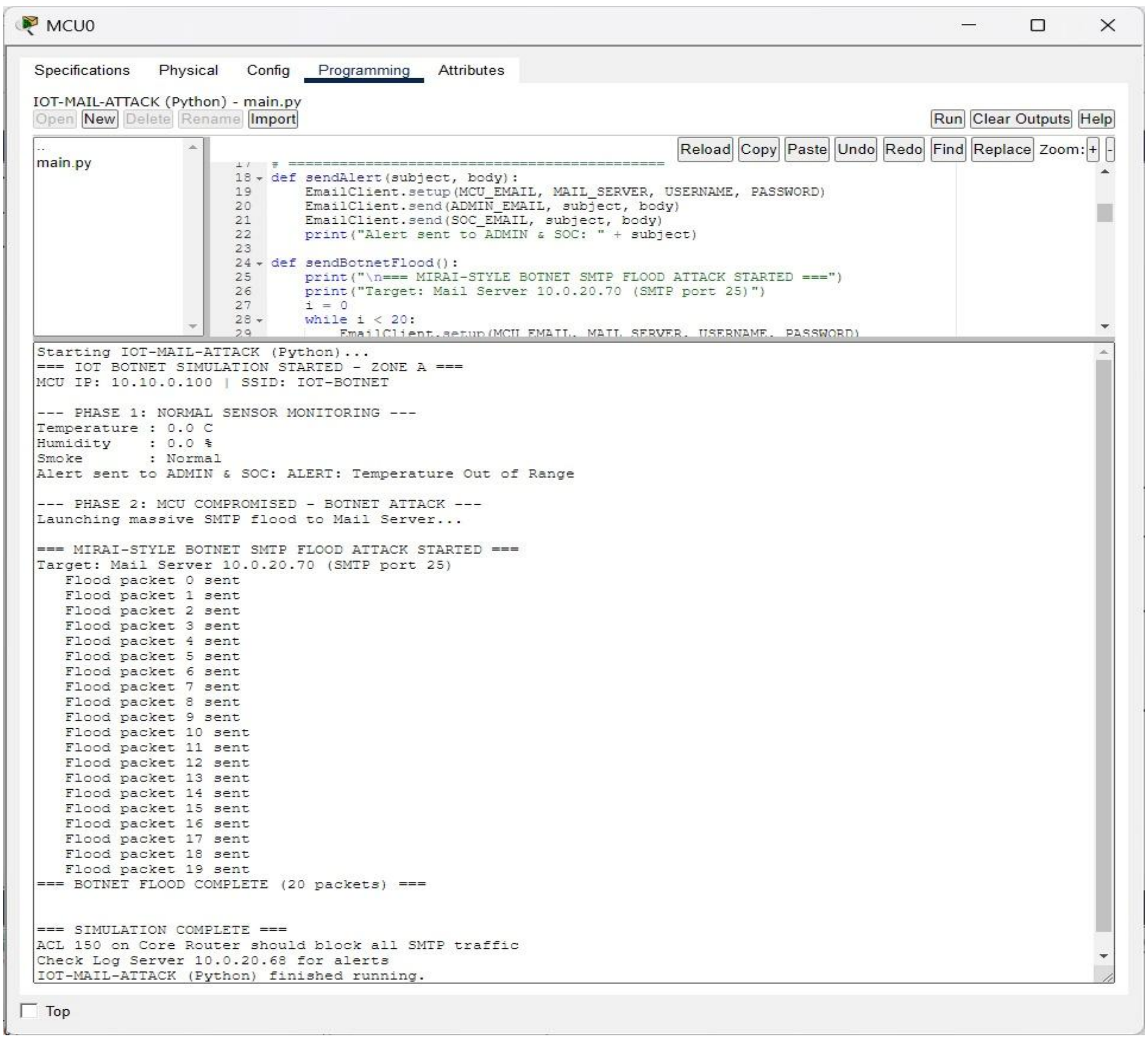

PDU Information at Device: Distribution Router-1
OSI Model
Outbound PDU Details
At Device: Distribution Router-1
Source: Distribution Router-1
Destination: Mail Server
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 10.0.20.70 ICMP Message Type: 8
Layer 2: Ethernet II Header 0060.3E70.9B78 >> 0004.9A4B.9449
Layer 1: Port(s): GigabitEthernet0/0/0
1. The Ping process starts the next ping request.
2. The Ping process creates an ICMP Echo Request message and sends it to the lower process.
3. The device encapsulates the data into an IP packet.
4. The device sets the TTL on the packet.
5. The device looks up the destination IP address in the CEF table.
6. The CEF table has an entry for the destination IP address.
Challenge Me
<< Previous Layer
Next Layer >>

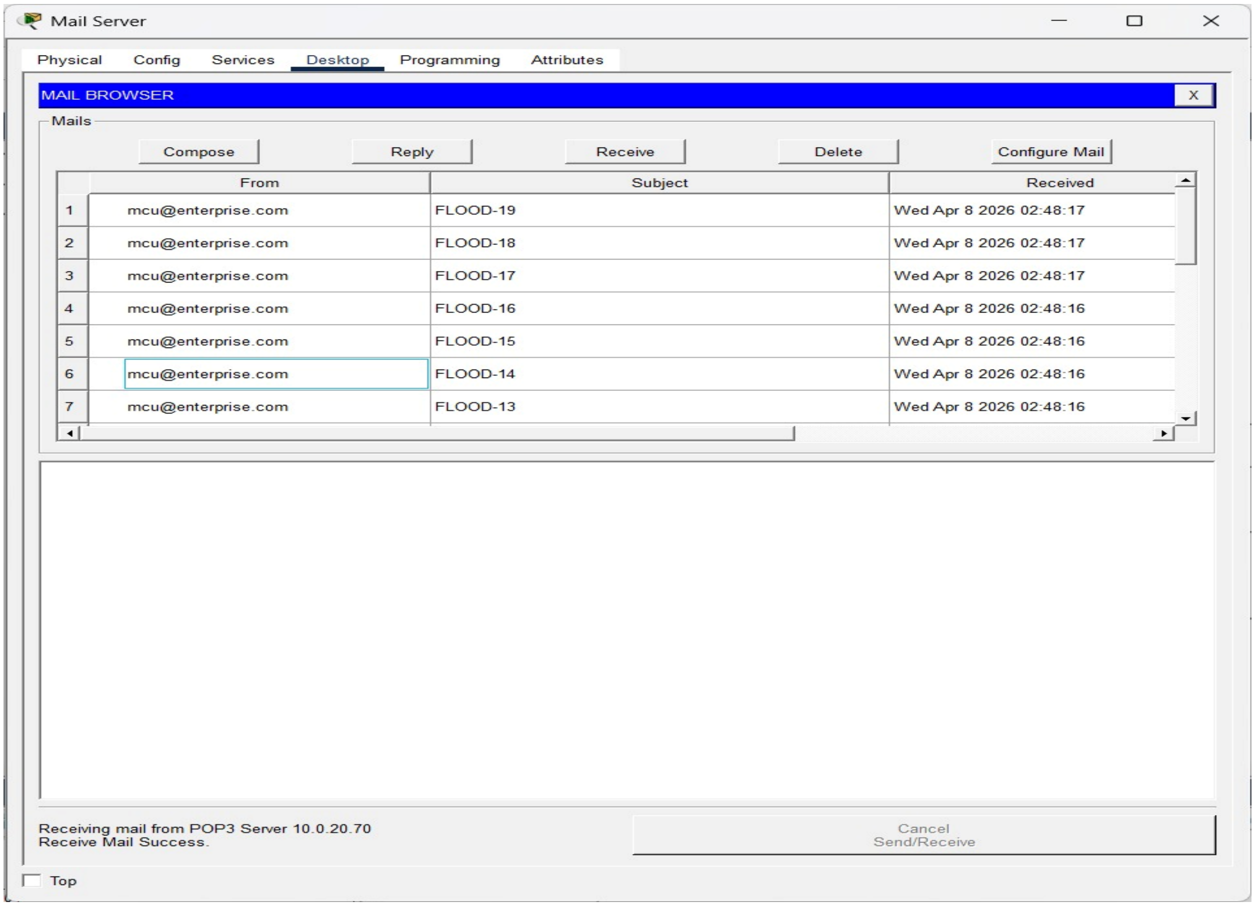

PDU Information at Device: NAT Router-1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: NAT Router-1
Source: Distribution Router-1
Destination: Mail Server
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 10.0.20.70 ICMP Message Type: 8
Layer 2: Ethernet II Header 0060.3E70.9B78 >> 0004.9A4B.9449
Layer 1: Port GigabitEthernet0/0/2
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 10.0.20.10, Dest. IP: 10.0.20.70 ICMP Message Type: 8
Layer 2: Ethernet II Header 0001.C96B.3AE4 >> 0002.4AEA.B2BD
Layer 1: Port(s): GigabitEthernet0/0/0
1. GigabitEthernet0/0/2 receives the frame.
Challenge Me
<< Previous Layer
Next Layer >>


PDU Information at Device: Mail Server
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: Mail Server
Source: Distribution Router-1
Destination: Mail Server
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 10.0.20.10, Dest. IP: 10.0.20.70 ICMP Message Type: 8
Layer 2: Ethernet II Header 0001.43A7.3101 >> 0001.63DE.391A
Layer 1: Port FastEthernet0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 10.0.20.70, Dest. IP: 10.0.20.10 ICMP Message Type: 0
Layer 2: Ethernet II Header 0001.63DE.391A >> 0001.43A7.3101
Layer 1: Port(s): FastEthernet0
1. FastEthernet0 receives the frame.
Challenge Me
<< Previous Layer
Next Layer >>

Network Address Translation: During communication from the Distribution Router network to the Mail Server, Network Address Translation (NAT) is used to modify the source IP address of outgoing packets. The original private IP address of the internal host is translated into the configured public or routable interface address of the NAT Router before forwarding the packet toward the destination server. This process hides internal addressing information, improves network security, and allows private networks to communicate with external resources using limited public IP addresses. The captured screenshots demonstrate the packet flow before and after translation, clearly showing the change in source IP address performed by the NAT Router. NAT also enables multiple internal devices to share a single external IP address through Port Address Translation (PAT), ensuring efficient IPv4 address utilization within the enterprise network.

Figure 10: MCU-Based IoT Security Simulation Showing Sensor Alerts & MTP Flood Attack

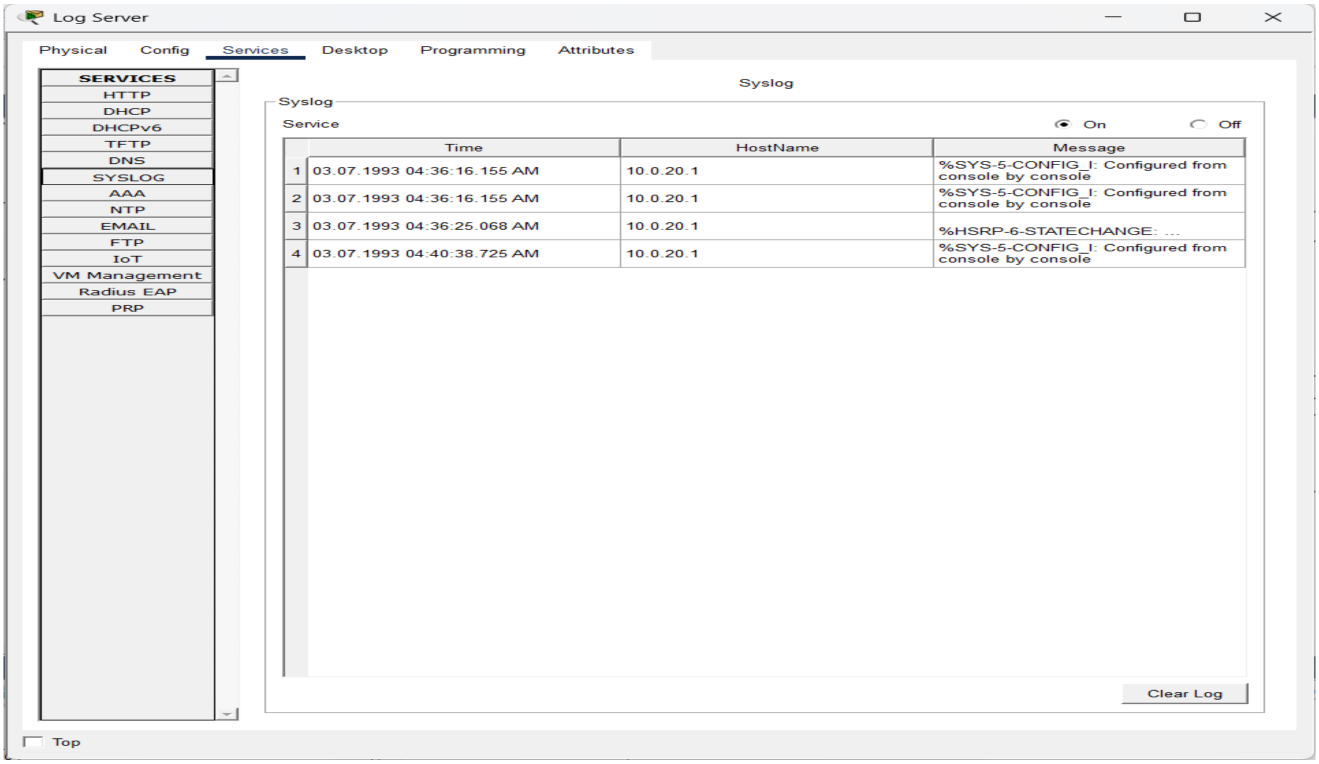




IoT MCU Security Simulation

An MCU board was integrated into the enterprise network to simulate an IoT endpoint device. The device was programmed using Python to monitor environmental parameters such as temperature, humidity, and smoke levels. Based on sensor readings, the MCU automatically generated email alerts to the network administrator and security operations center.To demonstrate cybersecurity threat scenarios, the MCU was also configured to simulate an IoT botnet-style attack by transmitting multiple SMTP flood packets toward the enterprise mail server. This was used to evaluate the effectiveness of implemented ACL security policies and centralized logging mechanisms. The simulation confirmed that unauthorized SMTP traffic could be identified and blocked by network access controls, while related events were successfully forwarded to the monitoring server for analysis and incident response.

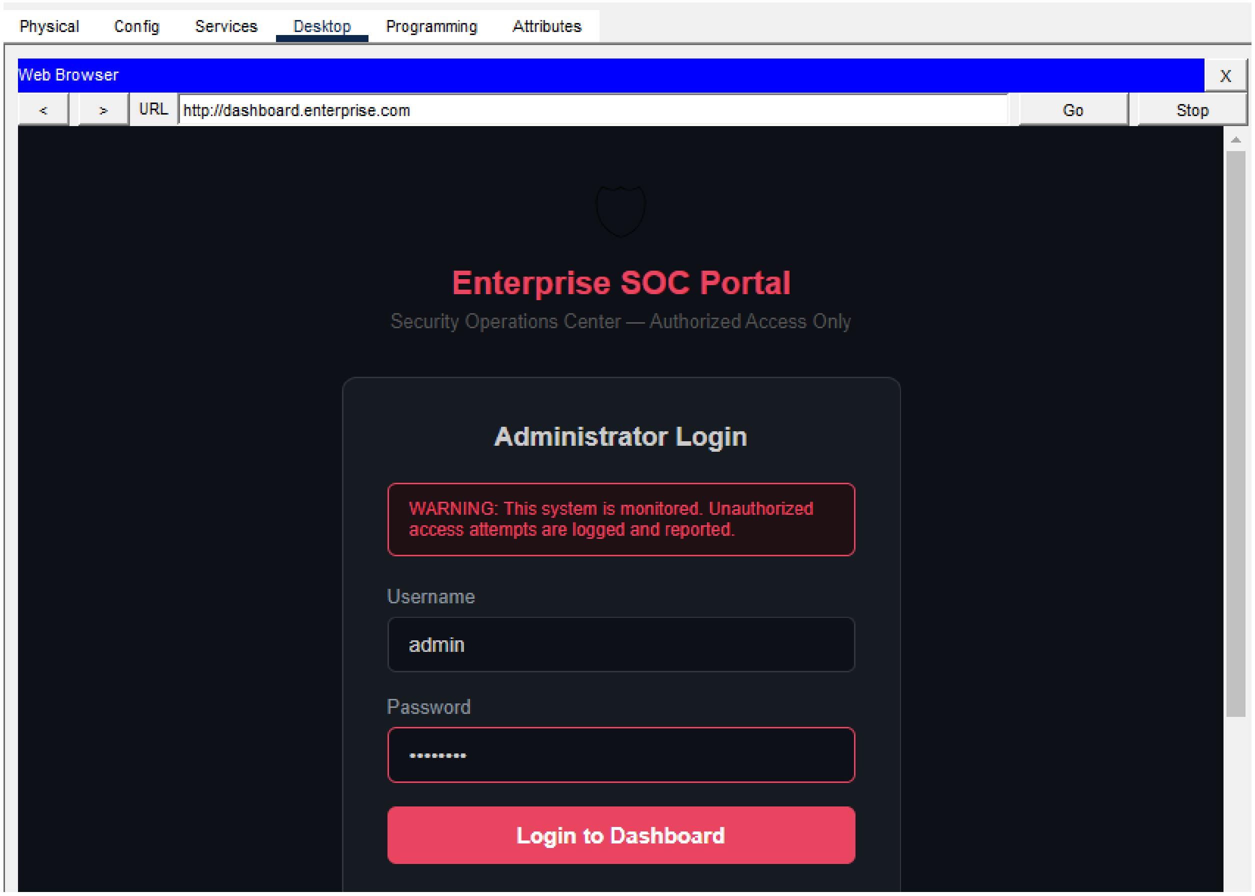
Figure 11: Working of Log Server



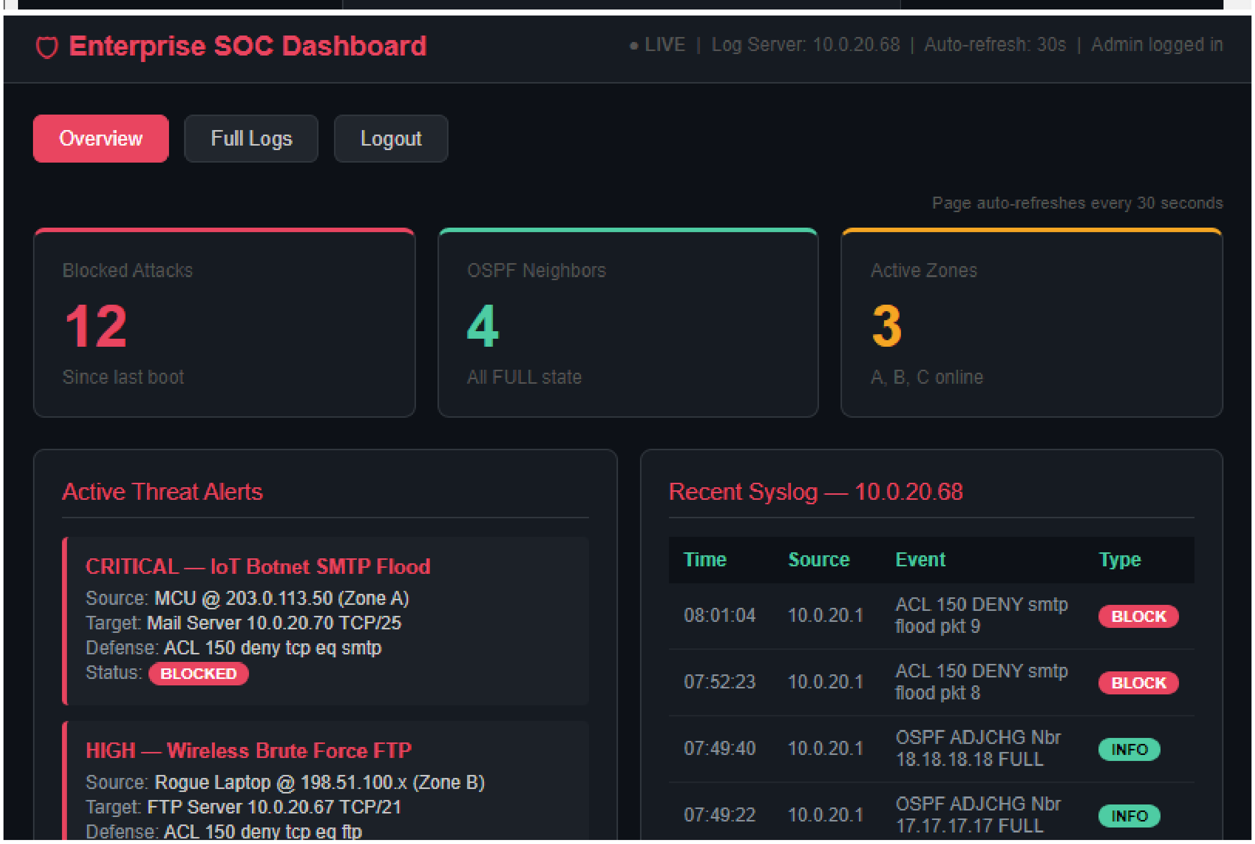
Log Server Mechanism

The Log Server was used as a centralized monitoring system to collect and store network events generated by routers, switches, and other connected devices. Syslog messages such as login attempts, interface status changes, routing updates, and security-related alerts were forwarded to the server in real time. This allowed continuous observation of network activity and helped identify faults, unauthorized access attempts, or abnormal traffic behavior. The centralized logging mechanism improved troubleshooting efficiency, security auditing, and overall network management.

Figure 12: FTP Server Web Dashboard with Authentication Interface



Physical
Config
Services
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://dashboard.enterprise.com
Go
Stop
Enterprise SOC Portal
Security Operations Center — Authorized Access Only
Administrator Login
WARNING: This system is monitored. Unauthorized access attempts are logged and reported.
Username
admin
Password
Login to Dashboard



Enterprise SOC Dashboard
● LIVE | Log Server: 10.0.20.68 | Auto-refresh: 30s | Admin logged in
Overview
Full Logs
Logout
Page auto-refreshes every 30 seconds
Blocked Attacks
12
Since last boot
OSPF Neighbors
4
All FULL state
Active Zones
3
A, B, C online
Active Threat Alerts
CRITICAL — IoT Botnet SMTP Flood
Source: MCU @ 203.0.113.50 (Zone A)
Target: Mail Server 10.0.20.70 TCP/25
Defense: ACL 150 deny tcp eq smtp
Status: BLOCKED
HIGH — Wireless Brute Force FTP
Source: Rogue Laptop @ 198.51.100.x (Zone B)
Target: FTP Server 10.0.20.67 TCP/21
Defense: ACL 150 deny tcp eq ftp
Recent Syslog — 10.0.20.68
Time
Source
Event
Type
08:01:04
10.0.20.1
ACL 150 DENY smtp flood pkt 9
BLOCK
07:52:23
10.0.20.1
ACL 150 DENY smtp flood pkt 8
BLOCK
07:49:40
10.0.20.1
OSPF ADJCHG Nbr 18.18.18.18 FULL
INFO
07:49:22
10.0.20.1
OSPF ADJCHG Nbr 17.17.17.17 FULL
INFO

5. Results & Discussion

After completing the configuration and implementation phase, a series of simulation tests were conducted in Cisco Packet Tracer to verify the functionality, reliability, and security of the proposed enterprise network. The results were evaluated based on connectivity, routing performance, redundancy operation, address translation, access control enforcement, wireless communication, and monitoring capabilities.

The screenshots and outputs presented in the implementation chapter provide practical evidence of successful device configuration and operational status. Therefore, this section focuses on the observed outcomes, performance analysis, and effectiveness of the implemented network design under different operational scenarios. Overall, the testing results confirmed that the network infrastructure operated according to the planned objectives and successfully supported secure, scalable, and reliable enterprise communication.

Figure 1: Before applying ACL to core router ping from IG3 side

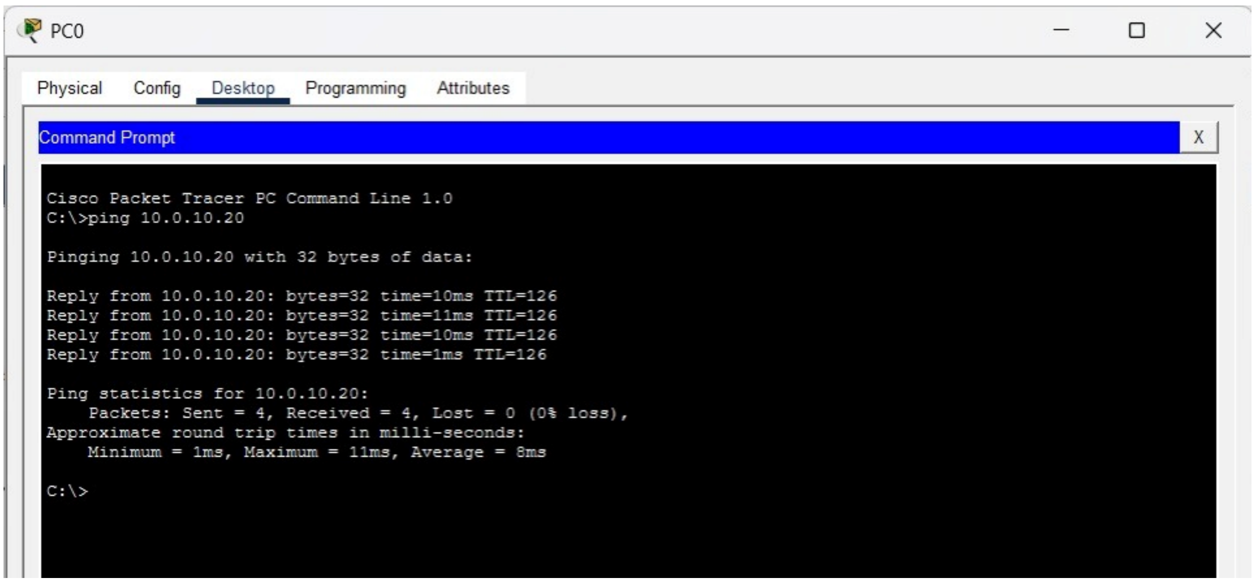


Figure 2: After applying the ACL on core router it blocks icmp packets and drop the ping request from IG3 sides

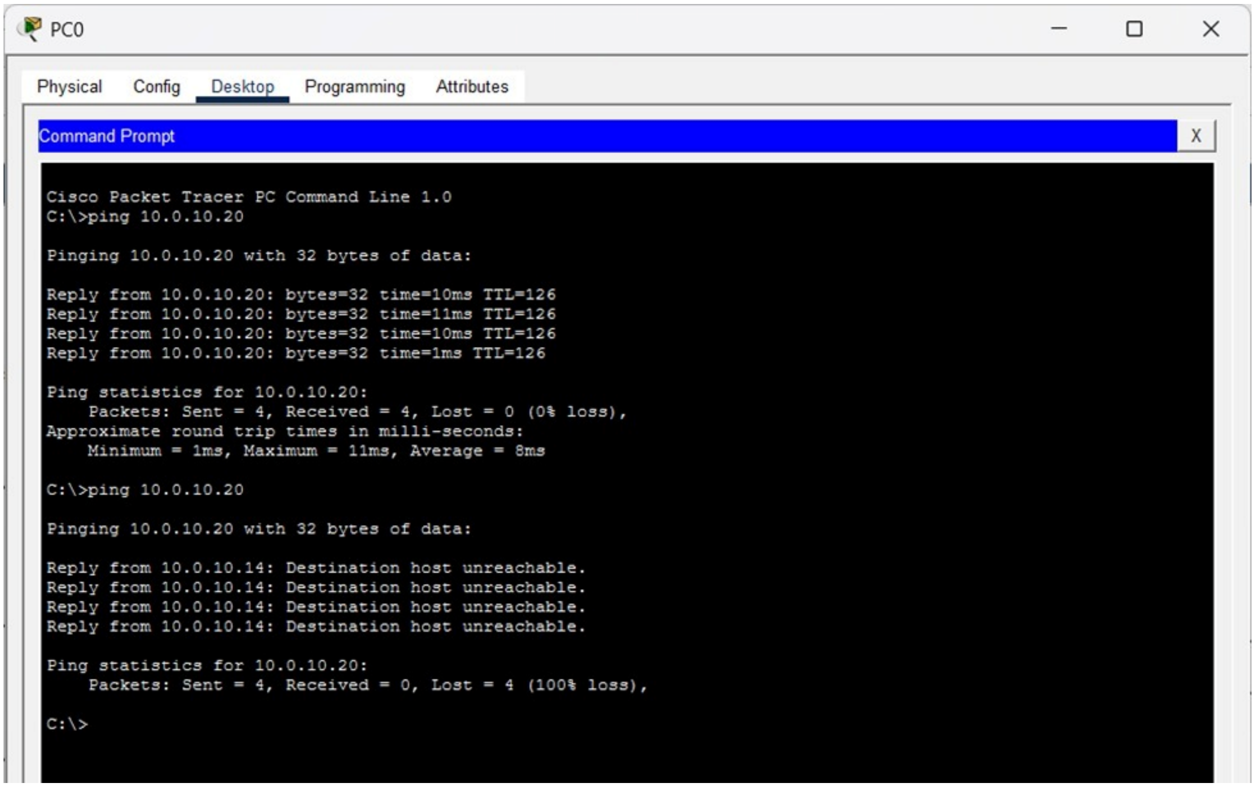
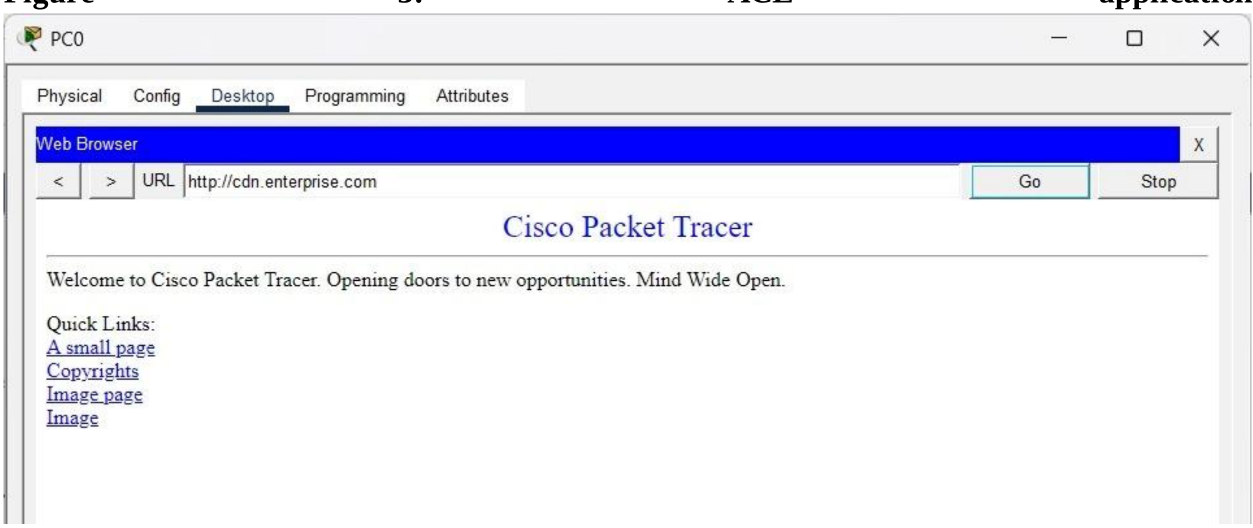
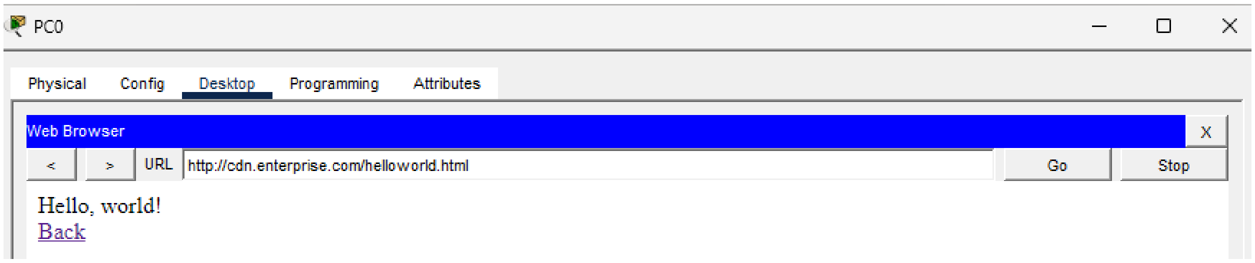


Figure 3: ACL application



Here, before applying the ACL we can browse the servers . Also after applying ACL it didn't block our TCP or http request we still can browse the website.



6. Limitations

Although the project successfully demonstrates a fully functional enterprise-grade network, certain limitations arise due to the simulation environment. Cisco Packet Tracer, being an educational tool, cannot replicate the exact performance, latency, or traffic volume of real-world hardware. The simulated IoT botnet attack and wireless intrusion scenarios were limited in scale and intensity compared to actual cyberattacks, making it impossible to fully stress-test the ACL 150, Syslog server, or routing convergence under extreme load.

The software also imposes constraints on advanced features such as deep packet inspection, intrusion prevention systems (IPS), integration with external SIEM solutions, or full-scale IPv6 deployment. Wireless capabilities are restricted to basic authentication methods, and the absence of live external internet connectivity limited the verification of certain NAT behaviors to internal traffic only. Finally, the large topology occasionally experienced simulation lag, affecting the smoothness of real-time testing.

These limitations are inherent to any Packet Tracer-based project and do not diminish the achievement of the stated objectives, but they highlight the gap between simulation and production environments. There are also some limitation of the logic and can be upgrade.

1 — Single Area OSPF: The entire network runs in OSPF Area 0 only. With 16 routers all in one area, every router maintains a complete link-state database of the entire topology. This increases CPU and memory usage on lower-priority routers like Distribution Routers which don't need full topology awareness. A multi-area design would significantly improve scalability.

2 — NAT Breaks End-to-End Visibility: The three NAT routers perform PAT overload hiding all nine Distribution Router networks behind a single IP. While this adds security, it means the server farm cannot initiate connections back to branch offices. Any server-to-branch communication is impossible without static NAT entries, which were not implemented.

3 — Static Default Routes on IG Routers: IG-Router-1, 2, and 3 rely on static default routes pointing to HSRP VIPs rather than learning defaults dynamically via OSPF `default-information originate`. If the HSRP VIP changes or a new core router is added, all three IG router static routes must be manually updated.

4 — No OSPF Authentication: OSPF runs without MD5 authentication on any adjacency. Any rogue router connected to the network with the correct area configuration could inject false LSAs and corrupt the routing table, potentially redirecting traffic through malicious paths.

5 — ACL 150 Applied Only on Core Router: ACL 150 is only enforced at Core Router's IIG-facing subinterfaces. If an attacker bypasses the IIG layer and enters from the Distribution side or directly accesses Core Switch, there is no ACL protecting the server farm from that direction. Defense in depth requires ACLs at multiple enforcement points.

6 — No Redundancy on Distribution Layer: Each of the nine Distribution Routers is a single point of failure for its branch subnet (172.16.x.0/24). If Dist-Router-2 fails, all devices on 172.16.2.0/24 lose connectivity with no failover mechanism. HSRP or dual uplinks were not implemented at this layer.

7 — SOC Dashboard is Static: The SOC dashboard hosted on FTP Server displays pre-populated log entries that do not update automatically from the live Log Server. In a real environment the dashboard would pull live Syslog data via API. The current implementation requires manual HTML updates to reflect new log entries.

8 — MCU Attack Simulation is Limited in Scale: The IoT botnet simulation sends only 10 SMTP flood packets due to Packet Tracer Python execution constraints. A real Mirai-style botnet generates hundreds of thousands of packets per second from thousands of compromised devices simultaneously. The current simulation demonstrates the concept but cannot stress-test the ACL or Syslog infrastructure under realistic attack volumes.

9 — No Link Redundancy on Core Switch Core Switch connects to Core Router and Backup Router via single FastEthernet links. If either physical link fails, OSPF reconverges but there is no EtherChannel or redundant path between these critical devices. A single cable failure between Core Switch and Core Router would cause a brief outage until OSPF reconverges.

10 — Wireless Security Limited to WPA-PSK The Home Router in Zone A uses WPA-PSK authentication for the IOT-BOTNET SSID. Enterprise environments use 802.1X with RADIUS authentication to provide per-device credentials and certificate-based authentication. WPA-PSK with a shared passphrase means any device knowing the password can join the wireless network.

7. Future Enhancements

1 — Multi-Area OSPF Implementation The current single-area OSPF design floods LSAs across all 16 routers unnecessarily. A future enhancement would divide the network into three areas — Area 0 as the backbone covering Core Router, Backup Router and Core Switch, Area 1 covering the NAT and Distribution Router layer, and Area 2 covering the IIG zone routers. This reduces LSA flooding, improves convergence speed, and allows route summarization at Area Border Routers, significantly reducing routing table size on all devices.

2 — OSPF MD5 Authentication To prevent rogue router injection attacks, MD5 authentication should be configured on all OSPF adjacencies. This ensures only authorized routers with the correct key can form neighbor relationships, protecting the routing domain from LSA spoofing and false route injection which the current design is vulnerable to.

3 — OSPF Route Summarization At the NAT router boundaries, the nine Distribution Router networks (172.16.1.0–172.16.9.0/24) could be summarized into a single advertisement (172.16.0.0/20) toward the core. Similarly the three zone subnets could be summarized at IG router boundaries. This would reduce routing table entries on Core Router and Core Switch from 16+ specific routes to 3–4 summary routes.

4 — HSRP on Distribution Layer Currently each Distribution Router is a single point of failure for its branch subnet. Adding a second Distribution Router per NAT group with HSRP configured between them would eliminate this vulnerability. If Dist-Router-2 fails, the standby router would take over the virtual gateway IP within seconds with no manual intervention required.

5 — Bidirectional NAT with Static Entries The current PAT overload implementation only allows outbound connections from Distribution networks. Adding static NAT entries for specific branch servers would allow the server farm to initiate connections back to branch resources, enabling centralized management, backup, and monitoring of branch devices which is currently impossible.

6 — Dynamic SOC Dashboard with Live Syslog Integration The current SOC dashboard displays static pre-populated log entries. A future implementation using a real web server with Node.js, Python Flask, or PHP could pull live entries directly from the Syslog server via UDP port 514 and display them in real time. Combined with WebSocket technology, new log entries would appear on the dashboard instantly without page refresh, creating a true live security monitoring interface.

7 — Full-Scale Botnet Simulation The current MCU Python script sends only 10 SMTP flood packets due to execution constraints. Migrating to GNS3 or EVE-NG with real IOS images would allow deployment of multiple attacking nodes simultaneously, generating realistic high-volume attack traffic to properly stress-test ACL 150 performance, Syslog throughput, and routing stability under load.

8 — 802.1X Wireless Authentication The IOT-BOTNET Home Router currently uses WPA-PSK shared key authentication. Replacing this with 802.1X RADIUS-based authentication would provide per-device credentials, certificate validation, and automatic session logging. This would prevent unauthorized devices from joining the wireless network even if the passphrase is compromised, and would integrate with the existing AAA infrastructure.

9 — EtherChannel on Core Links The single FastEthernet links between Core Switch and both Core Router and Backup Router represent physical single points of failure. Implementing 802.3ad LACP EtherChannel with two parallel links per connection would provide both redundancy and doubled bandwidth on the most critical links in the topology.

10 — SNMPv3 and NetFlow Monitoring Adding SNMPv3 polling on all 16 routers and Core Switch would enable real-time interface utilization, CPU, and memory monitoring from a central NMS. Combined with NetFlow export on Core Router and NAT routers, traffic patterns and anomalies could be detected proactively before they escalate into outages or security incidents, replacing the current reactive Syslog-only approach.

8. Conclusion

This project successfully designed, implemented, and tested an Enterprise-Grade Secure Network Architecture with Active Threat Defense using Cisco Packet Tracer. By integrating OSPF for dynamic routing, HSRP for gateway redundancy, NAT for address translation and security, extended ACLs for traffic filtering, and a centralized Syslog server for monitoring, the network achieved high availability, scalability, segmentation, and proactive threat defense — all core requirements of modern enterprise environments.

The simulation of real-world attack scenarios (IoT botnet SMTP flooding on the Mail Server and unauthorized wireless access attempts on the FTP Server) demonstrated that the implemented security controls effectively detected, blocked, and logged malicious activity, validating the layered defense approach.

Through this project, critical enterprise networking concepts — hierarchical design, dynamic routing convergence, first-hop redundancy, address management, and integrated security — were applied in a single cohesive architecture. The experience gained provides strong practical insight into the challenges and best practices of designing resilient, secure, and scalable networks.

Ultimately, this work serves as a realistic model for how contemporary organizations can protect their digital infrastructure against failures and cyberattacks while maintaining continuous business operations. The knowledge and skills acquired are directly transferable to industry environments and lay a solid foundation for further studies in advanced networking and cybersecurity.

9.Knowledge Profile Coverage (K1–K8)

Knowledge Category	Coverage in Project
K1: Depth of Knowledge	Applies advanced concepts including OSPF, HSRP, NAT/PAT, extended ACLs, 802.1Q VLANs, and Syslog - all integrated within a 22-device enterprise topology.
K2: Knowledge of Disciplines	Combines computer networking, cybersecurity, and IoT simulation in a single coherent design.
K3: Knowledge of Engineering Practice	Follows industry standards: three-layer hierarchical design, SSH-only management, PortFast on edge ports, and DHCP relay for centralized addressing.
K4: Research & Investigation	Reviewed enterprise design models and security architectures to identify gaps in simulation-based redundancy and threat defense, guiding protocol selection.
K5: Problem Analysis	Diagnosed OSPF convergence conflicts, HSRP failover timing, ACL rule ordering, and NAT translation issues using show commands and systematic verification.
K6: Design & Development	Designed a 22-device three-tier hierarchical network with HSRP redundancy, NAT/PAT boundaries, VLAN segmentation, ACL-enforced server farm, and a subnet-restricted SOC monitoring dashboard.
K7: Engineering Tools & Techniques	Used Cisco Packet Tracer for simulation; applied structured IP addressing, VLAN planning, subnetting, and ACL policy design as core engineering techniques.
K8: Social & Environmental Impact	Promotes secure, reliable digital infrastructure that protects data privacy, reduces downtime risk, and supports business continuity.

10.Complex Engineering Problem Attributes (P1-P7)

Attribute	Project Alignment
P1: Depth of Knowledge	Simultaneously applies OSPF, HSRP, NAT, extended ACLs, VLAN segmentation, Syslog, and IoT simulation — requiring deep, cross-disciplinary technical knowledge to integrate and troubleshoot.
P2: Wide-Ranging or Conflicting Issues	Security ACLs must not block legitimate traffic; OSPF convergence must not disrupt HSRP failover; NAT must preserve host connectivity. Balancing these across 22 devices involves inherently conflicting design constraints.
P3: Abstract Thinking & Analysis	Planning logical segmentation, subnetting backbone links, deciding ACL placement, and anticipating protocol interactions before implementation required significant abstract reasoning and pre-design analysis.
P4: Infrequently Encountered Issues	Integrated MCU Python botnet simulation, ACL threat blocking, Syslog correlation, and JavaScript-authenticated SOC dashboard within a single enterprise topology.
P5: Adherence to Standards	Strictly follows OSPF (RFC 2328), HSRP (RFC 2281), 802.1Q, SMTP/IMAP, SSH, and Syslog (RFC 5424) -ensuring protocol compliance, interoperability, and professional engineering practice.
P6: Stakeholder Conflicts	User zones, server administrators, IoT operators, and security teams have different access requirements. Multiple independent ACL policies resolve these conflicts while preserving normal operations.
P7: Interdependent Sub-Problems	OSPF convergence affects HSRP failover; ACL rules influence NAT flow; VLAN design determines Syslog reachability. A misconfiguration in any one subsystem can cascade across the entire network.